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RESEARCH PAPER

Laser-based surface pretreatment for ultrasonic metal welding of copper and dissimilar aluminium-copper joints

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Abstract

Ultrasonic metal welding (USMW) is a widely used solid-state welding process for electrical components. Despite careful dimensioning, defective welds occur due to the large number of influencing variables, which sometimes remain undetected by existing monitoring strategies. One known major influencing factor is the surface condition of the components to be welded, which is controlled in industrial applications using mostly chemical or mechanical cleaning methods. Existing surface-based mechanisms in USMW are insufficiently described in the literature or contradict each other, so that corresponding cleaning methods are usually developed on an application-specific basis. To further complicate efforts to define an optimal process, the surfaces of the materials used are usually only standardized in general terms, but not for USMW. This work, therefore, investigates the suitability of the surfaces of commonly used copper and aluminium materials for USMW and then defines an optimal state. Based on this, a nanosecond-pulsed laser process is developed that best meets these requirements. In addition, the potential of optimizing the welding process of copper joints to the laser-treated surface condition to achieve a robust overall process chain and transfer the results to the specific requirements of aluminum-copper joints is demonstrated. The welded joints are characterized in detail mechanically, electrically and metallographically in order to quantify the surface influence on the welding result.

Keywords Ultrasonic metal welding · Laser cleaning · Surface treatment · Copper welding · Aluminium-copper welding

Abbreviations

Al	Aluminium
I	Current
Nd:YAG	Neodymium-doped yttrium aluminum garnet
USMW	Ultrasonic metal welding
Cu	Copper
GD-OES	Glow discharge spectroscopy
PMT	Photomultiplier optics
UV-Vis	Ultraviolet-visible spectroscopy
ETP	Electrolytic tough pitch
HV	Vickers hardness
U	Voltage
XPS	X-ray photoelectron microscopy

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1 Introduction

USMW is used in industry for a wide range of applications for the energy and mobility transition, such as power electronics and battery cell production [1–3]. The materials used in this area are primarily pure and low-alloyed copper and aluminium materials. Despite the metallurgical challenges, Al-Cu dissimilar joints are increasingly being used in this area to reduce costs and component weight, for which USMW is ideally suited. The short process duration and the characteristics of a solid-state welding process with low energy input ensure high economic efficiency [4, 5]. At the same time, the process is characterized by a large number of influencing and disturbance variables that must be kept under control in order to enable a robust process [1, 4, 6, 7] These include, for example the component geometry [8], the material hardness [9]or the surface condition [10] of the components to be welded. In addition to the base materials used, these influencing variables ensure specific but characteristic process behaviour [4, 6, 9, 11–13].

1.1 Challenges in USMW of similar and dissimilar joints for electrical applications

Müller et al. [8] have shown that the geometric component properties and the base material itself have the greatest impact on a successful USMW when the component-related influencing variables are considered separately. This result is also consistent with the studies of other scientists [11, 14]. However, the optimization of the component geometry for a USMW process is often limited due to other technical and installation space requirements, so that simple adjustments to radii or bores are often used [15] in order to make the components suitable for USMW.

USMW is also frequently used in partially or fully automated processes in which geometric boundary conditions can already be kept relatively well under control due to the fixture design and precise positioning masks. The cleanliness and surface quality, on the other hand, are often not tightly regulated, are usually not checked before the welding process and can vary considerably when changing batches or suppliers. This is caused by the fact that industrially used materials are usually subject to multi-stage production and transport processes that lead to the formation of a characteristic composition of a metallic surface (see Fig. 1). The undisturbed base material only forms the surface after electrochemical polishing. Even minor impurities or changes to the surface structure (oxides, cleanliness and roughness) can significantly affect the USMW process, resulting in increased scattering of the joint strength, welding time or even component deformation [6, 10, 11]. Certain coatings, e.g. with tin, even make the USMW almost impossible [1]. Accordingly, coatings, impurities and scattering of the surface quality are often considered undesirable [11, 16], as joint formation only takes place when the surfaces to be joined approach each other at the atomic level and bare metal surfaces can come into contact. The removability of surface contamination must therefore be taken into account [11]. To a limited extent, these layers can vaporize or break up during the cleaning phase of the welding process and be transported out of the joining zone [17]. Harthoorn observed significantly weaker bonds in oily samples and those with soft oxide layers [11], as these reduce friction in the joining zone. Müller [7, 12], on the other hand, can demonstrate improved joint strength in some cases with slight surface contamination with diluted rolling oils. Hard oxide layers are usually less critical [4]. Nunes [10] also states that mechanical processing of the surface of electrolytic tough pitch cold-rolled copper used commercially for USMW is unnecessary, as this reduces the joint strength.

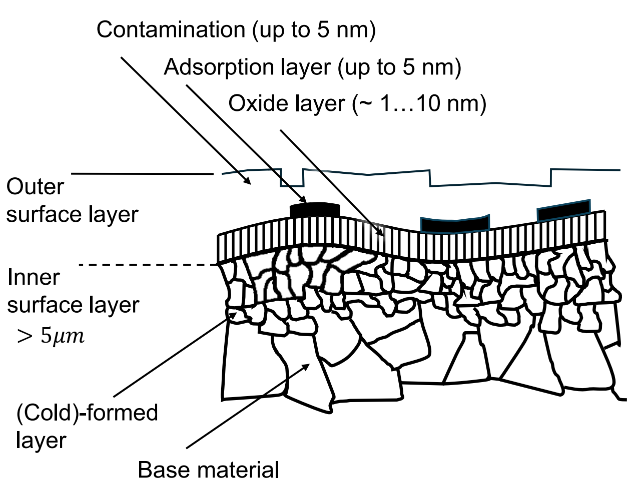

Fig. 1 Real, processed material surface according to Wodara et al. [1]

Corresponding knowledge or basic know-how about these influencing variables is still lacking, contradicting or only given qualitatively or phenomenologically [4, 6]. Even if the process itself can remove some contamination, cleaned surfaces generally ensure better reproducibility of the welding quality, as surface treatment creates a uniform initial state [18].

Technologies for industrial surface monitoring to ensure the initial state are also often still poorly represented in USMW. With regard to the influence of the surface (oxide layer, roughness, cleanliness), there is a close relationship to bonding, soldering and adhesive processes, in which, on the other hand, there is already an understanding of the importance of the component surface and basic knowledge about the influence on joint formation [19–23]. In the case of USMW, this has so far only existed to a limited extent. The demand for fast, industry-compatible methods for the in-line characterization of surface properties such as texture and roughness (e.g. by gloss measurement [12, 24]), cleanliness (e.g. by fluorescence measurement [25]) or oxide layer thickness (e.g. by UV–Vis [26, 27]) for joining processes is consequently high.

1.2 Effect of material oxidation on USMW

Like any metal, aluminium forms a natural oxide layer in contact with oxygen, the growth of which usually comes to a standstill after a few nanometers and protects the underlying material from further corrosion. The naturally formed oxide layer of pure aluminium usually has a thickness of between 1 and 3 nm (dry environment) and 4–10 nm (humid conditions) [22, 28]. The oxide layer on aluminium usually consists of two parts: the water-containing and porous top layer (e.g. $Al(OH)_3$) and an almost pore-free barrier layer containing only Al_2O_3 [29]. Other foreign atoms such as carbon or hydrogen can also be adsorbed or embedded in the porous top layer. A temperature of up to 300 °C in a dry environment has no effect on the natural oxide layer (1–3 nm)

[22], but if there is relevant humidity or condensation, the oxide layer can continue to grow even at room temperature [28]. The barrier layer forms within one hour in dry air [30].

First, a thin layer of cuprite (copper(I)-oxide or Cu_2O) [26, 31] forms on bare copper under atmospheric conditions. With further increasing oxidation, a copper(II)-oxide layer forms at the air–material interface. [32] In practice, copper(I)-oxide and copper(II)-oxide thus occur together. The growth of an oxide layer on bare copper is highly dependent on the ambient conditions. In a pure oxygen atmosphere (dry air, 25 °C), an oxide layer ~ 1.5-nm thick forms within a few hours, which is then largely stable and only grows very slowly [33–35]. Already at temperatures of 50 °C [33], in the presence of humidity or oxidative air components [35–37], the oxide layer can grow steadily.

Watanabe et al. [5] have investigated the influence of the oxide layer on the USMW of Al-Cu joints. A lower joint strength is reached with growing oxide layers, whereby the oxide layer on the copper surface has a significantly greater influence on joint formation and strength than the aluminium surface. An increase in the Cu oxide layer from ~1 to ~4 nm already reduces the joint strength by around 15%, and with an oxide layer of ~21 nm the joint strength drops by around 75%. Accordingly, an influence on the joint strength is also to be expected for copper-copper joints, even if quantified investigations are lacking.

1.3 Conventional and laser-based surface cleaning

Filmic impurities or organic surface residues are generally unevenly distributed [19, 25]. The roughness and oxide layer thickness are also not uniform as a result of production, transport and storage processes. The introduced irregularities (oxide layers, cleanliness and roughness) are undesirable with regard to a robust and efficient welding process, even if they can be compensated at the expense of process capability to a certain level [4, 16].

According to DIN 8592 [38], the achievable degree of cleanliness of a surface depends on the cleaning process and the type and nature of the contaminants. For example, surfaces with a higher degree of roughness are usually more difficult to clean. Cleaning surfaces by means of chemical-industrial methods is often not an optimal solution, even after multi-stage processes. Only laboratory cleaning guarantees almost residue-free surfaces [19]. In addition, the chemical processes used are usually comparatively harmful to the environment and difficult to use locally [39, 40].

Pulsed laser systems for surface cleaning are already being used successfully in industry for local paint stripping or preparing components for bonding or soldering processes [41, 42]. The laser offers the potential for precise, contactless and residue-free cleaning as well as localized processing of the surface topography (roughness and texture) [43]. Under perfect conditions, the laser system works wear-free and remains largely independent of the degree of contamination compared to chemical cleaning. This means that adsorbed OH- or H-C compounds in the porous top layer (aluminium) or the oxide layer itself can be removed. If the parameters are optimized, the base material remains largely unaffected [44, 45]. If the base material is melted locally without inert gas cover or vacuum, however, (re-)oxidation of the surface can be expected [46]. Laser processing of bare copper and aluminium surfaces, which are frequently used in electrical applications, is challenging with the solid-state lasers in the infrared range (~ 1000-nm wavelength, pulsed operation in the nanosecond range) commonly used for cleaning surfaces. Less than 5% (copper) or 7% (aluminium) of the focused laser irradiation is absorbed [47, 48]. Furthermore, the degree of absorption depends on the angle of incidence, the maximum pulse intensity and the surface condition. Matt surfaces or a higher roughness, for example, lead to improved absorption. In the case of copper in particular, absorption increases sharply as soon as a melt is present, which can lead to an unstable cleaning process [49].

The influence of laser-based cleaning and micromachining of copper and aluminium surfaces on USMW has not yet been fully investigated, while initial investigations already show possible advantages [50, 51]. The introduction and acceptance as a cleaning process for USMW require careful investigations into existing interactions and the determination of process limits.

1.4 XPS and GD-OES for surface analysis

Analyzing the influence of the surface on the USMW process and adjusting defined surface properties with the laser requires detailed knowledge of the surface composition. Laboratory methods such as X-ray photoelectron microscopy (XPS) or glow discharge spectroscopy (GD-OES) are usually used for detailed analysis of the chemical surface composition of metallic materials such as copper or aluminium [52]. XPS is characterized by the fact that not only the elemental composition, but also chemical bonding states can be investigated. The depth of information is usually limited to ~10 nm [53]. The depth of information can be increased by combined etching (e.g. using argon) and subsequent XPS measurement. At the same time, the lateral resolution is in the micrometer range. The procedure is generally time-consuming and the interpretation of the results is challenging. GD-OES is characterized by the fast measurement of thicker layers [52], as it works with high etching rates. At the same time, the lateral resolution is in the millimeter range. The determination of bonding states, on the other hand, is difficult. The methods are unsuitable for series testing during production and serve as a starting point for the development of fast, industry-compatible methods.

1.5 Summary and research objective

Ultrasonic metal welding is considered to be particularly surface-sensitive. Generally valid and quantified basic knowledge on the influence of disturbance variables (cleanliness, roughness, oxide layer), which is already taken for granted in other joining processes such as soldering or bonding, is still lacking. Studies carried out by various scientists on the influence of surfaces on the USMW contradict each other or differ greatly [1, 9–11, 50]. This can be attributed to the fact that even the smallest changes or differences in surface condition have a significant influence on the welding result. Necessary laboratory analysis methods such as XPS to analyze these differences are expensive and time-consuming, meaning that such a detailed analysis is rarely carried out. Furthermore, the influence of cleanliness, roughness and oxide layer has mostly been examined separately from each other, although a significant influence of these disturbance variables on each other is to be expected. A comparability of the investigations in literature is thus only partially given. The advantage of laser treatment is also the challenge: it simultaneously influences cleanliness, chemical composition, texture and roughness. Thus, these elements should also be considered in combination. This also comes much closer to real-life applications.

The objective of this work is to describe and determine surface-related influencing variables on USMW quantitatively. Laboratory methods (XPS/GD-OES) and industry-standard methods such as fluorescence and gloss measurement are used for this purpose. Based on this, an optimal condition in terms of roughness, cleanliness and oxide layer is defined, which serves as the target value for laser processing. Subsequently, a laser processing procedure is developed for both aluminium and copper, which addresses the material-specific requirements. In addition, the necessary properties of a laser treatment for industrial use (e.g. increase in the coupling limit intensity due to contamination) are determined to achieve uniform surface conditions. Finally, it is shown that an efficient overall process chain is possible if USMW is optimized to a laser-processed state. Similar copper and dissimilar aluminium-copper joints are used to investigate relationships and transferability of the results.

2 Methodology and experimental setup

At the beginning of this study, the materials and sample geometries used are presented and characterized with regard to their mechanical–technological properties. Based on this, the methods used for sample preparation (chemical/mechanical/laser-based), the welding setup for USMW and the strategies for analyzing the results are presented.

2.1 Materials and analysis techniques

In this study, an electrolytic tough pitch cold-rolled copper strip in hard-rolled condition—Cu-ETP-R290 [54]—with 1 mm material thickness and cold-rolled aluminium in semi-hard-rolled condition—EN AW-1050-H14—with 1.5-mm thickness is used. The material is tested mechanically in terms of strength (tensile test) and hardness (Vickers hardness) (Table 1).

The material surface is characterized in terms of roughness, cleanliness and gloss. Oxide layers and chemical binding states are also analyzed on a sample selection basis via XPS and GD-OES. The surface roughness S_a and line roughness R_a (mean arithmetic height according to DIN 25178 [56]) are determined optically using a laser confocal microscope (Keyence VK-X-1000). The gloss is determined in accordance with ISO 2813 [24] as gloss unit (GU) at measuring angles of 20°, 60° and 85° using a combined instrument (PCE-PGM 100-ICA), whereby the evaluation in this work is limited to the generally valid measuring angle of 60°. Cleanliness is tested using fluorescence measurement (RFU) with an industrially used hand-held device (SITA Cleanospector).

A Spectruma GDA 750 HR with photomultiplier optics (PMT) and 4 mm anode diameter in pulsed operation (800 V, 777 Hz, 8% duty cycle) is used for GD-OES. The process is pressure-controlled and argon is used as carrier gas at a pressure of 2.5 hPa (Al) or 3.0 hPa (Cu). The measurement time is 140 s (Al) or respectively 60 s (Cu) for a ~ 1000-nm depth profile. The real measuring range on the sample is approximately 5 mm in diameter, so that the measured values are averaged over this range. The copper is examined in the chemically cleaned (*U*) and oxidized state (T180)—see Table 2—and aluminium in the chemically cleaned (*I*)—see Table 3—state.

The ESCALAB 250Xi device (ThermoFisher Scientific), including the MAGCIS double-beam ion source, is used for XPS. Overview images of the surfaces (survey scans), detailed analyses of the binding states as well as depth profiles down to the base material (depth profiling) are carried out. The X-ray spot size is 400 µm in diameter in each case. Sputtering was carried out with a spot diameter of 1.5 mm (mono and cluster) at an incident angle of 30°. A 75 Ar + ion cluster with an acceleration voltage of 8 kV was used for copper sputtering to analyze the material in the depth profile. The sputtering rate for a 75 Ar + ion cluster on SiO_2 with the device used is ~ 0.42 nm/min [57]. The sputtering rate of SiO2 using monoatomic Ar + ions at 2 kV acceleration voltage (used for aluminium sputtering)

Table 1 Material properties of samples ($n=8$) [55]

Mechanical properties	Tensile strength (R_m) [MPa]	Yield strength ($R_{p0,2}$) [MPa]	Stretch (A_{50}) [%]	Hardness (HV 0.1) [-]	Roughness (R_a) [µm]
Factory certificate (Cu-ETP-R290)	310	303	12	102	0.34
Measured value in rolling direction (Cu-ETP-R290)	299	-	-	102	0.39
Factory certificate (EN AW-1050-H14)	116	111	9	-	-
Measured value in rolling direction (EN AW-1050-H14)	106	-	-	39	0.58

was determined ~ 8–9 nm/min. The conversion of sputtering rates for the used materials and their oxides is carried out on the basis of literature values. The materials copper and aluminium are each examined in four states (uncleaned (*R*)/chemically cleaned (*U*)/laser 1 (L1)/laser 2 (L2)).

2.2 Specimen and surface preparation

The samples are cut from the strip (Cu) or from a sheet (Al) using mechanical cutting machines. The upper sample has the dimensions 45 × 20 × 1 mm (L × W × H); the lower sample has the dimensions 80 × 40 × 1 mm (Cu) or 80 × 40 × 1.5 mm (Al).

2.2.1 Cu-Cu-similar joints

In addition to a reference state (*R*), 14 surface variations were carried out, resulting in a total of 15 test series (Cu-Cu)—see Table 2. Six of the present conditions were created using laser processing. In the reference state (*R*), the sheets are used, ex works with possible contamination (rolling oil, dust, particles). The chemically cleaned samples (*U*) are cleaned using 2-propanol by wiping a soaked cloth over the top and bottom of the sheets. All other samples are first chemically cleaned on both sides and then further processed, whereby the further processing is only done on the sides facing each other. Group O (OL and OH) is contaminated with 30 µl diluted rolling oil (1 oil:1000 isopropanol resp. 1 oil:100 isopropanol). Group G (GL, GH and GO) is mechanically processed with abrasive fleece (SiC-P600-1000, Al_2O_3-P180-220) and GO is then contaminated with 30 µl diluted rolling oil (1:100). The thermally aged samples (T180) are stored in a climate chamber at 180 °C for 3 days. The lectropol-5 device (Struers GmbH) is used for electropolishing (*E*). The samples are not processed in any other way beforehand. The electrolyte used is "D2"—a Struers formulation consisting of ethanol, phosphoric acid, 2-propanol, urea and distilled water. The

Table 2 Sample overview of weld specimen (Cu-Cu similar joints)

Test series	Laser parameter/additional information	Abrasive material [grain size]	Cleaning	Contamination Diluted rolling oil	Specimen
R	Reference (uncleaned)	-	-	EXW	10
U	Chemically cleaned	-	2-propanol	-	10
OL	Contamination	-	2-propanol	1:1000	8
OH	Contamination	-	2-propanol	1:100	8
GL	Grinding	P600–1000	2-propanol	-	8
GH	Grinding	P180–220	2-propanol	-	8
GO	Grinding and contamination	P180–220	2-propanol	1:100	8
T180	Thermally aged (180°C, 3 d)	-	2-propanol	-	8
E	Electropolishing	-	2-propanol	-	8
L1	Laser cleaning (L1)	-	2-propanol	-	10
L2	Laser processing (L2)	-	2-propanol	-	10
OLL1	Laser cleaning (L1)	-	2-propanol	1:1000	5
OLL2	Laser processing (L2)	-	2-propanol	1:1000	5
OHL1	Laser cleaning (L1)	-	2-propanol	1:100	5
OHL2	Laser processing (L2)	-	2-propanol	1:100	5

Table 3 Sample overview of weld specimen (Al-Cu-dissimilar joints)

Test series	Laser parameter / Additional information	Abrasive material [Grain size]	Cleaning	Contamination Diluted rolling oil	Speci-men
R	Reference (uncleaned)	-	-	EXW	8
I	Chemically cleaned	-	2-Propanol	-	8
OL	Contamination	-	2-Propanol	1:100	8
OH	Contamination	-	2-Propanol	1:50	8
GL	Grinding	P600 - 1000	2-Propanol	-	8
GH	Grinding	P180 - 220	2-Propanol	-	8
LA	Al: yes (L2), Cu: no	-	2-Propanol	-	8
L	Al: yes (L2), Cu: yes (L2)	-	2-Propanol	-	8
LC	Al: no, Cu: yes (L2)	-	2-Propanol	-	8

processing parameters are summarized in Table 4. After electropolishing, the samples are rinsed with ethanol and dried with a hot air dryer. Electropolishing with the used parameters removes ~ 20 µm of the surface so that the base material is fully exposed.

Of the six laser-processed groups, three are processed with the parameter set for cleaning (L1) and three for remelting (L2). Groups OXXX (OLL1, OLL2, OHL1, OHL2) are contaminated with diluted rolling oil prior to laser processing in the same way as group O.

In addition, welding parameter studies are carried out with laser-machined Cu sheets (L1 and L2) in order to optimize the welding process for the new surface condition and to identify a robust overall process chain. For this purpose, full-factorial test plans (design of experiment) with two levels (low, high) and three factors (force, amplitude and welding energy) are created and welded. The target value of the optimization is the maximum peel tensile force in the 90° peel test. Sample specimens are analyzed metallographically.

2.2.2 Al-Cu-dissimilar joints

In addition to a reference state (*R*), eight surface variations are carried out, resulting in a total of nine test series (Al-Cu). Three of the existing conditions are created using laser processing. In the reference state (*R*), the sheets are used, ex works with possible contamination (rolling oil, dust, particles, etc.). The chemically cleaned samples (*I*) are cleaned using 2-propanol by wiping a soaked cloth over the top and bottom of the sheets. All other samples are first chemically cleaned on both sides and then further processed. Group O (OL and OH) is contaminated with 30 µl diluted rolling oil (1 oil:100 isopropanol, resp. 1 oil:50 isopropanol). Group G (GL and GH) is mechanically processed with abrasive fleece (SiC-P600-1000, Al_2O_3-P180-220). In the laser-processed group, both sheets (*L*) and only one sheet (LA, LC), i.e. only copper or only aluminium, are processed. Eight samples are prepared in each case. After welding, all samples are electrically tested using 4-wire measurement; five samples are then subjected to a mechanical test (90° peel tensile test), and two samples are analyzed by micrograph (microscopy). One sample remains for subsequent geometric measurement. The copper sample is placed on the upper side, and the aluminium sheet on the lower side. The sample overview for Al-Cu dissimilar joints can be found in Table 3.

2.2.3 Laser processing

A CL20 system (max. average power 20 W) from Clean-Lasersysteme GmbH is used for laser processing. This equipment is selected due to widespread use, its simple scalability to higher power, and comparatively low acquisition costs of pulsed solid-state lasers in the infrared range, as this promises the highest industrial relevance. The system operates in pulsed mode (*Q*-switch) with a pulse duration of $t = 100$ ns and is operated exclusively in the focused position. A detailed specification is summarized in Table 5. The diode-pumped solid-state laser (Nd:YAG) operates at a wavelength of 1064 ± 4 nm with a Gaussian beam profile. The parameter range for laser processing is determined using the method presented in Helfers et al. [12].

Example surfaces in the target parameter range for copper and aluminium (Table 6) are then laser processed, and the

Table 4 Electropolishing processing parameters (Struers lectropol-5)

Elektrolyte	Voltage (U) Voltage [V]	Current (I) Current [A]	Area (A) Bearbeitete Fläche [cm^2]	Time (t) Cycle time [s]	Tempera-ture (T) [°C]
D2*	24	~ 5.2—5.4	2	20	~ 25

Table 5 Fixed laser parameters and specification values for CL20 (workpiece level)

Parameter	Value
Max. mean laser power (specified/measured)	20 W/15.91 W
Pulse length	100 ns
Pulse frequency	40–200 kHz
Focus length f	160 mm
Scan speed	0.05–3 m/s
Track/pulse overlap	50%
Spot diameter 1/e^2	60 µm
Max. pulse intensity	~ 140 MW/cm^2
Max. fluence	~ 14 J/cm^2
Max. pulse energy (specified/measured)	0.5 mJ/~ 0.4 mJ

procedure is checked. A comparison of laser system specifications (Table 5) and the required pulse intensities for bright copper (Table 6) shows that initial vaporization is not possible, as the maximum pulse intensity is not sufficient. As soon as the surface is melted or if there is a pulse overlap, vaporization with the system is also possible.

Due to the significant differences in the material properties relevant for laser treatment between aluminium and copper, these are listed in detail in each case. Literature values for the solid base material are given as absorption coefficients A [59]. These values change abruptly as soon as a melt is present. The significantly higher difference between the melting and evaporation temperatures for aluminium means that it is easier to produce a stable melt without high evaporation during laser processing.

Based on the stitch tests presented in the results section, a parameter set for cleaning (L1) and remelting (L2) is defined for copper and aluminium respectively (Table 7) and used for further welding experiments. Parameter set L1 is selected in order to improve the cleanliness of the surface (e.g. oil residues) without changing the texture, while parameter set L2 processes both texture and roughness as well as cleanliness through complete remelting. Processing is carried out in a meandering manner perpendicular to the rolling direction.

2.3 Welding setup and mechanical–electrical test procedure

The welding experiments were conducted on a Schunk LS-C ultrasonic welding machine with 4 kW generator power and a pneumatic cylinder with 80-mm diameter. The linear oscillating system, consisting of the associated converter, 1:1 booster and a λ/2-horn with a working area of 5 × 5 mm and diagonal, pyramidal roughening (0.5 mm × 90°), has a working frequency of ~ 20 kHz. The anvil is also roughened with 0.5 mm × 90°. The weld specimens are placed overlapped in a mask made of glass-bead-filled UHMWPE—cf. Müller et al. [13]—with increased wear resistance (Fig. 3a). The upper specimen is inserted loosely into the fixture; the lower specimen is clamped with a toggle clamp. The tests are carried out with statistically optimized welding parameters according to Table 8 (Design of Experiments—DoE) and optimized specimen geometry with respect to the avoidance of interfering natural frequencies (see Fig. 2b). An energy-controlled process is used. The amplitude of 27 µm is achieved in (undamped) idle mode. By means of a full

Table 6 Pulse intensity for laser processing (t = 100 ns, ρ_{Cu}=8960 kg/m^3, c_{p_Cu}=385 J/kgK, ρ_{Al}=2700 kg/m^3, c_{p_Al}=896 J/kgK) [47, 48, 58]

Pulse intensity calculation	Temperature $T(x=0,t)$ [K]	Thermal conductivity (λ_{th}) [W/mK]	Absorption range (A) [%]	Th. penetration depth (l_{th}) [µm]	Pulse intensity range (I) [MW/cm^2]
Melting (Cu)	1356.15	390–400	2.4–4.8	6.72–6.82	92–187
Vaporization (Cu)	2868.15				194–396
Melting (Al)	933.35	230–240	3.3–6	6.17–6.3	23–43
Vaporization (Al)	2743.15				94–179

Table 7 Variable laser parameters (calculated fluence and intensity values on specimen surface)

Parameters	Cu cleaning (L1)	Cu remelting (L2)	Al cleaning (L1)	Al remelting (L2)
Pulse frequency [kHz]	40	40	93	93
Laser power [%]	70 (10.3 W)	100 (15.9 W)	40 (4.6 W)	80 (12.2 W)
Pulse intensity [MW/cm^2]	~ 90	~ 140	~ 17	~ 46
Pulse fluence [J/cm^2]	9	14	1.7	4.6
Pulse energy [mJ]	0.26	0.4	0.05	0.13
Surface rate [cm^2/s]	0.35	0.35	0.88	0.88
Scan speed [m/s]	1.2	1.2	3	3

Table 8 Welding parameters

Welding parameters	Cu-Cu	Al-Cu
Force [N]	1500 (3 bar)	750 (1.5 bar)
Amplitude [µm]	27 (100%)	27 (100%)
Energy [Ws]	800	1000

factorial test plan, the joint of the uncleaned condition (*R*) was optimized for a maximum peel force in the 90° peel pull test. The process was deliberately not optimized to a cleaned surface condition. If the welding parameters are optimized for a cleaned surface condition, organic contamination (e.g. rolling oil) can lead to considerable overwelding. In order to consider the influence of the surface on the ultrasonic welding process in as isolated as manner as possible, the boundary conditions that the sheets do not fall apart or are even completely stamped through before a quality parameter is determined are therefore included as an additional optimization criterion. Two exemplary welding samples with laser-processed surfaces are shown in Fig. 2a.

Compared to a conventional USMW system, the welding system is equipped with additional sensors and high-frequency data acquisition (500 kS/s) per channel. In addition to the internal system data, the electrical generator power, the indentation depth (LVDT sensor), and the oscillation amplitude at the horn and anvil are recorded. The amplitudes are measured via laser triangulation sensors (LTS). Laser processing and welding experiments are performed the same day to avoid re-contamination or unwanted natural oxidation. The mechanical test is carried out using a standard tensile testing machine (Zwick Z5.0—max. 5 kN tensile force) and a test fixture specifically manufactured for the application (Fig. 3b). The electrical resistance of the welded samples is measured using 4-wire measurement at a measuring distance of 30 mm across the welding zone with a current of 50 A for 10 s.

3 Results and discussion

The results of laser-based surface treatment, surface characterization and welding tests are presented below. The results are then correlated with peel strength, macrosections and electrical contact resistance before being subsequently discussed.

3.1 Laser processing

Calculations in the methods section show that completely different processing parameters are required for copper and aluminium. In addition, spot tests on the coupling threshold on copper (Fig. 4) indicate that even a slight contamination of the surface causes a significant increase in the coupling threshold. At the same time, these validation tests confirm that exact knowledge of laser system specifications and the used materials is essential for precise processing.

Figure 5 shows an example of the laser processing of the materials used in the uncleaned state with the parameters specified for the test series. For both materials, but especially for copper as a result of the technical deviations in the surface condition, even with the laser parameter L1, localized coupling into the surface and associated melting occurs (Fig. 5c and d). Due to the principle and material (Gaussian profile, high-energy pulse, highly reflective

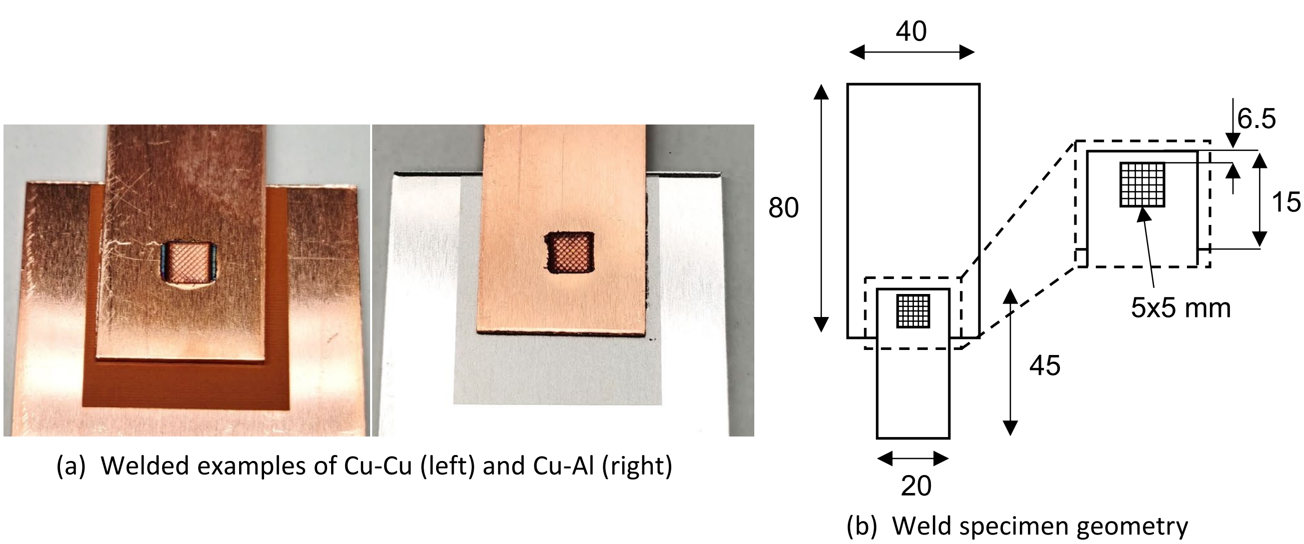

Fig. 2 Welded examples and specimen geometry

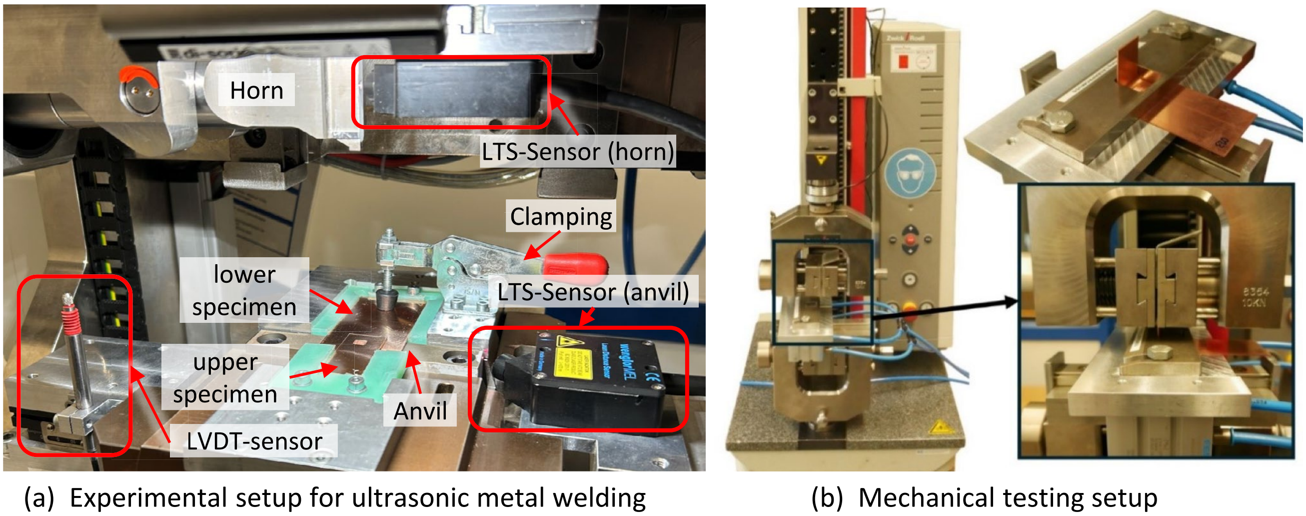

Fig. 3 Welding and mechanical testing setup

materials), pronounced cratering occurs even with processing parameters that are only slightly above the coupling limit. Accordingly, an increase in surface roughness is to be expected. In addition to craters, laser processing of the copper results in more pronounced spatter formation. In the case of aluminium, the individual craters have approximately the theoretical laser spot diameter of ~ 60 µm, and the formation of splashes is comparatively low (Fig. 5e). On the copper surface, however, the craters have a diameter of only ~ 40 µm and are overlaid by many copper splashes (Fig. 5f).

The parameter set for cleaning (L1) leads to the complete removal of the 1:1000 oil dilution; the higher contamination (1:100) is already incompletely removed. A comparison with the investigations from Helfers et al. [12] shows that a higher pulse intensity and a higher pulse overlap (> 50%) can also be effective in removing organic contaminants without remelting the surface. Multiple processing is also possible, but is not investigated further here.

3.1.1 XPS

To analyze the surface composition, an uncleaned sample (*R*) and a cleaned sample (*U*) of copper are examined using XPS survey (Fig. 6a and b). Apart from the expected peaks for oxygen (O1s), copper (Cu2p, Cu3p and CuLmm) and carbon (C1s), no other elements can be determined on the cleaned sample. Other peaks in the spectrum that arise as

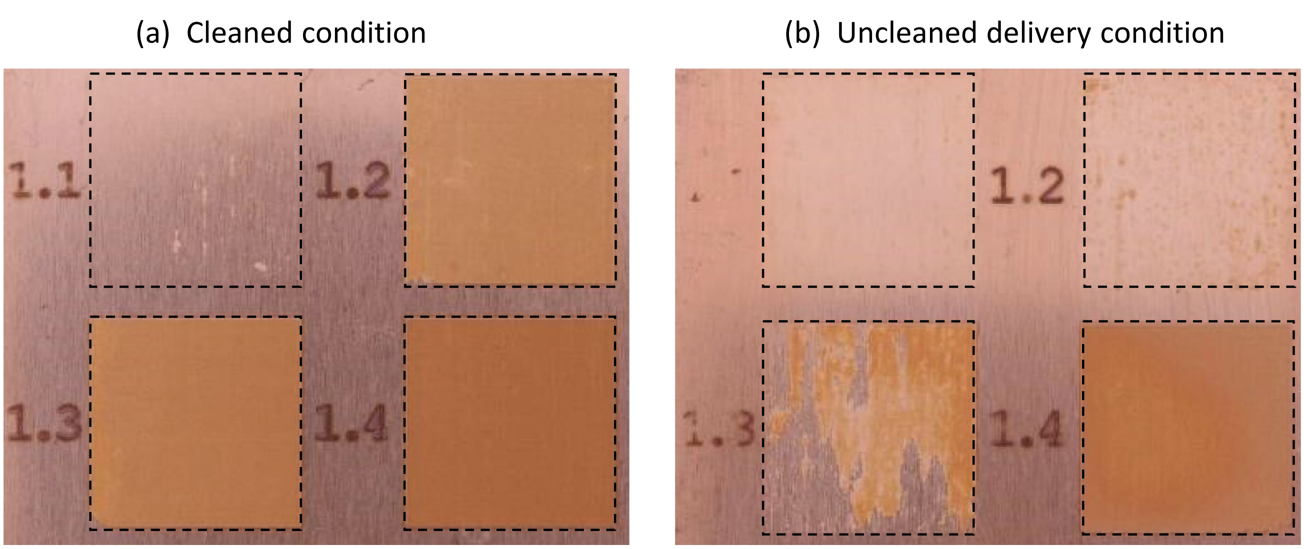

Fig. 4 Comparison of the laser coupling threshold (melt) on Cu-ETP R290 with 70% (1.1), 75% (1.2), 80% (1.3) and 85% (1.4) laser power

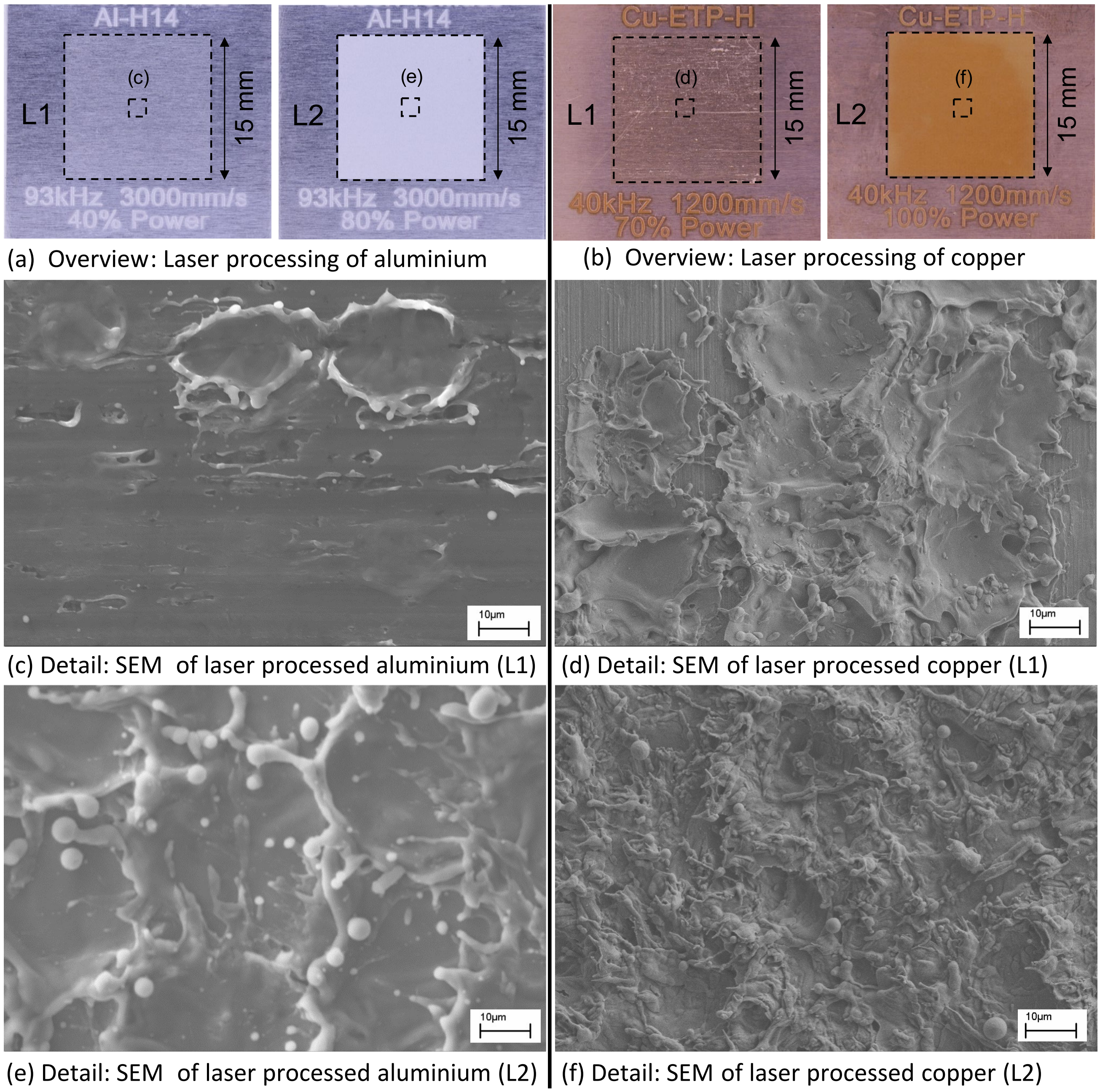

Fig. 5 Image overview and detailed scanning electron microscope images (SEM) of laser processing

secondary peaks of the elements mentioned are also marked in the XPS survey. In addition, a depth profile is carried out using an argon cluster gun (Ar-75) for 120 s. This examination also shows no unexpected elements. A more pronounced C1s peak (carbon) and a small Si2s peak (silicon) are recognizable on the uncleaned sample, which indicates particulate or dust contamination. Both the cleaned and the uncleaned Cu sheet show adventitious carbon contamination in the C1s peak (usually min. 1–2 nm thickness if samples were previously exposed to atmospheric conditions). Apart from the adventitious carbon on the surface, the material used is of such high quality straight from the factory that a deep surface removal seems unnecessary.

The rolling process during production therefore did not introduce any near-surface impurities into the material. The exact quantification of the oxide layer or the proportions (Cu, Cu_2O, CuO) from the XPS survey is only possible by analyzing the Auger spectral lines (LMM). Alternatively, this can be visualized via the XPS depth profile using an argon cluster gun.

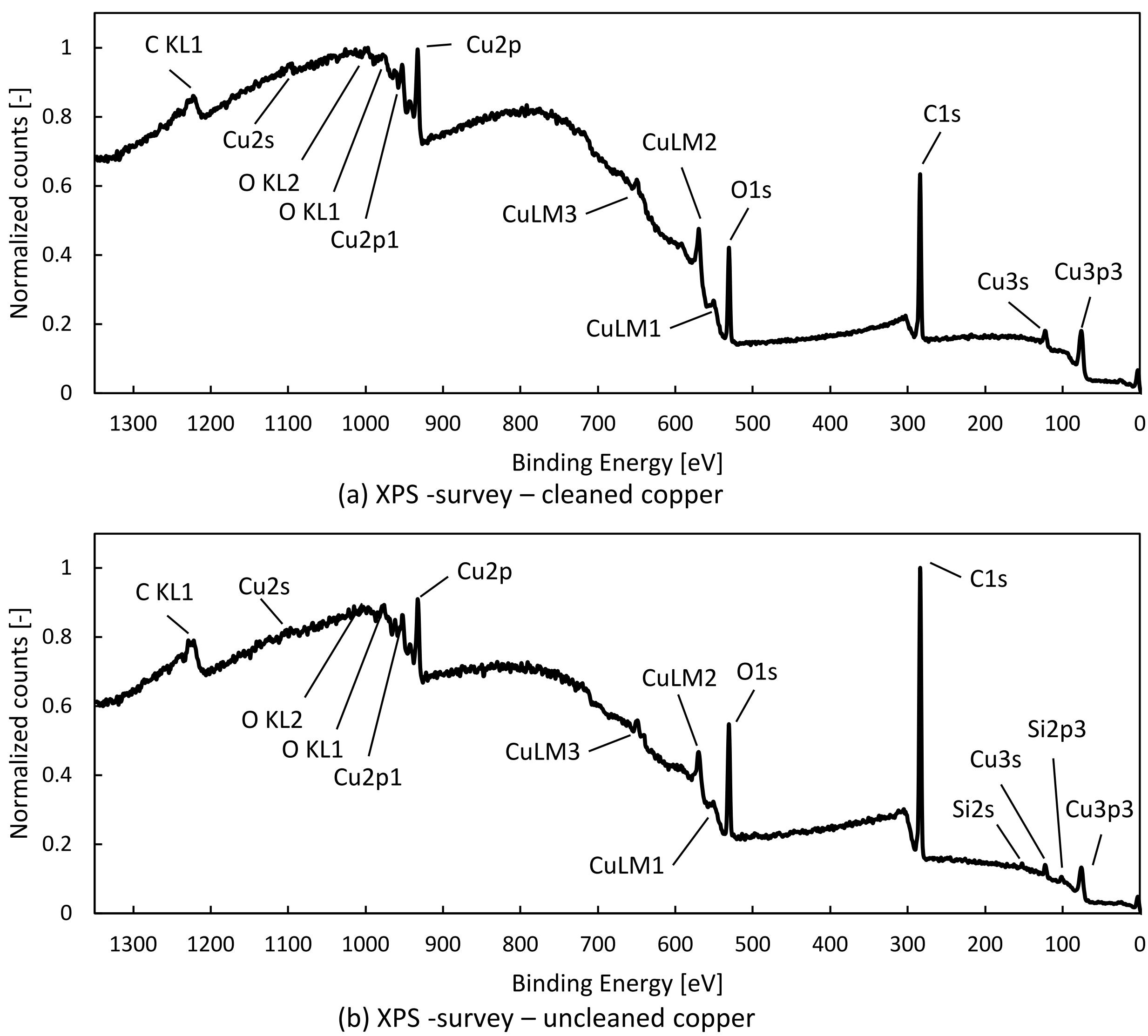

Fig. 6 XPS-survey comparison for Cu-ETP R290 (without baseline correction)

A detailed examination of the binding energies, counts, and peak positions for copper (Cu2p), oxygen (O1s) and carbon (C1s) allows some conclusions to be drawn about the surface state of reference and laser-processed samples (Fig. 7a–h). In the cleaned state, both metallic copper Cu(0) and a small proportion of Cu(I) or Cu(II) oxide can be determined by correlation with reference spectra [60]. The unpurified sample has a narrower main Cu2p peak at ~ 932.6 eV and less pronounced twins. This is followed by a thinner oxide layer than in the purified sample. To quantify the oxide layer, argon sputtering is used instead of the extended analysis of the Auger parameters (LMM). The correlation of the results shows oxide layers of < 10 nm for all samples, even without sputtering. The C1s peak for all samples shows the usual pattern of a, metallic surface that has been exposed to the atmosphere. Analyzing the O1s peaks is difficult without further information but shows that bound oxygen is present on the surface of all samples. The correlation of the laser-processed and cleaned states so far only allows an estimation of the oxide layer thickness, as precise sputter rates for the present materials and surface composition are missing. Assuming an equivalent sputtering rate of 0.42 nm/min (75 Ar + ion cluster) for copper oxides comparing to SiO_2 and that the base material is reached after ~ 100–180 s (O1s peak intensity decreased by half), the oxide layer thickness can be estimated to ≤ 1.5 nm for the cleaned and uncleaned states. Laser parameter 1 (L1) leads to a slight increase in the oxide layer. For laser parameter 2 (L2), the Cu2p peak shows significantly more pronounced twins. The correlation with reference spectra shows a clearly increased Cu(II) oxide layer on the surface [60]. The oxide layer thickness remains less than 10 nm

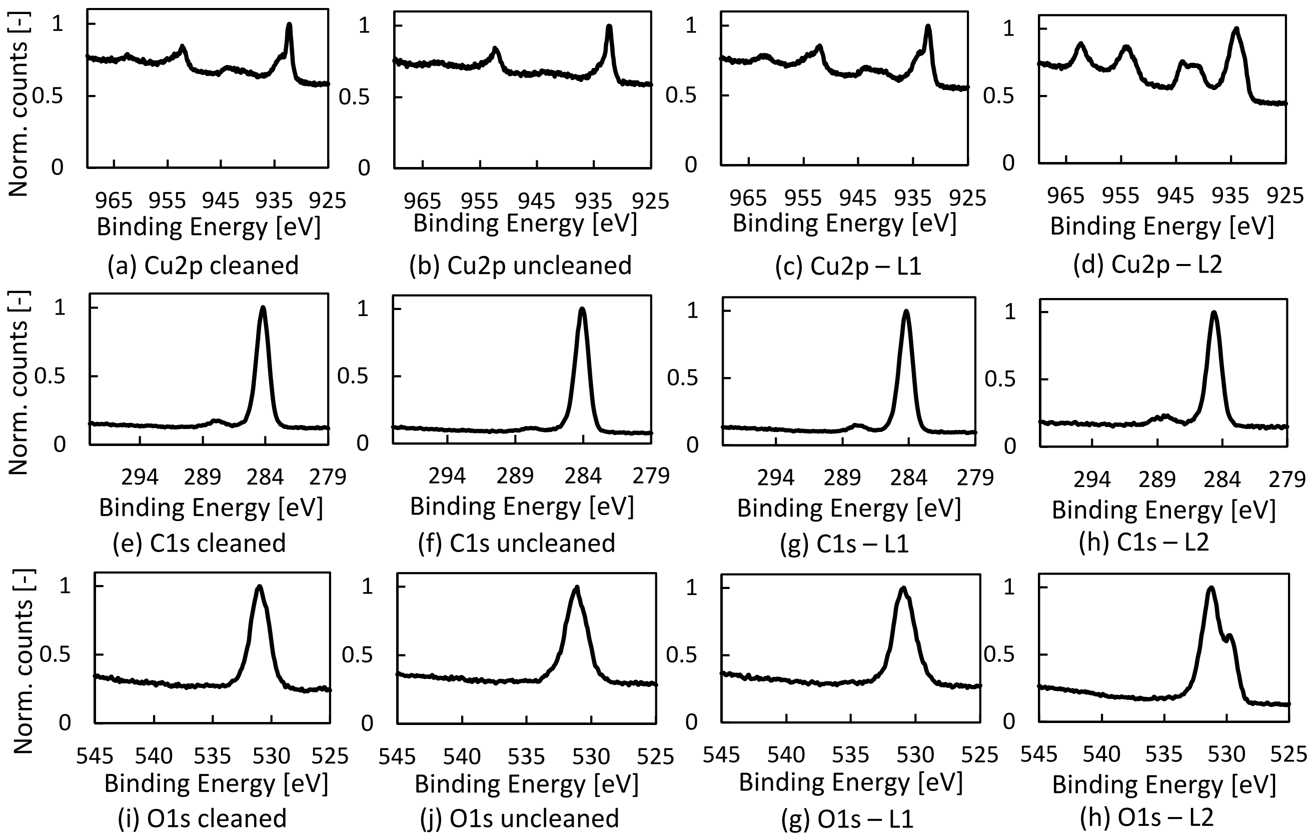

Fig. 7 Initial XPS-spectra for Cu2p, C1s, and O1s of cleaned, uncleaned and laser-processed Cu-ETP R290 samples

overall. The pronounced satellite in the O1s peak at ~ 529 eV as a result of laser processing with L2 (Fig. 7h) confirms the formation of oxide. After ion etching the laser 2 sample for 360 s (75 Ar + ion cluster), the O1s peak is still not decreased by half. This leads to a minimum thickness of the oxide layer of ~ 2.5 nm after laser parameter 2. In addition, the C1s intensity remains at around 2/3 of the initial level even after 360 s of ion-etching, meaning that some of the adventitious carbon is transported from the surface to deeper layers. For the other three samples, however, the C1s intensity decreases continuously with the sputtering process.

Due to the high surface roughness ($R_a > 1$ µm) and the formation of craters, XPS measurements of the laser-machined surfaces only represent a very local investigation. The size of the XPS spot diameter is a multiple of a single crater (Fig. 5).

The results of the XPS measurements on the untreated copper materials are consistent with those from the literature on bare copper materials [33] with regard to the oxide layer thicknesses determined.

Equivalently, an uncleaned aluminum sheet (*R*) and a cleaned aluminum sheet (*I*) were examined using XPS-survey (Fig. 8). Apart from the expected peaks for oxygen (O1s), aluminium (Al2p and Al2s) and carbon (C1s), no other elements can be determined on both samples. Since the uncleaned aluminum sheets are provided with a polyethylene protective film, it is not surprising that less contamination is detected compared to the uncleaned copper sheets. The rolling process during production, therefore, did not introduce any near-surface impurities into the Al material.

A detailed examination of the binding energies, counts and peak positions for aluminium (Al2p), oxygen (O1s) and carbon (C1s) allows some conclusions to be drawn about the surface state of reference and laser-processed samples (Fig. 9). All Al specimens show adventitious carbon contamination in the C1s peak. In comparison to Zähr et al. [29], no peak shift in the O1s peak can be detected to differentiate between oxides and hydroxides on the surface of the investigated samples. At the time of the XPS measurement, a natural oxide layer had already formed again on the laser-processed samples. The Al2p region already shows the usual peaks for metallic aluminum (~ 72.6 eV), aluminum oxides and hydroxides (~ 74.1–75.2 eV) [61] on the cleaned and uncleaned sheet in the initial spectrum. The laser-processed sheets do not show this distinction. Only in the sputter depth profile do the characteristic peaks also become visible in the

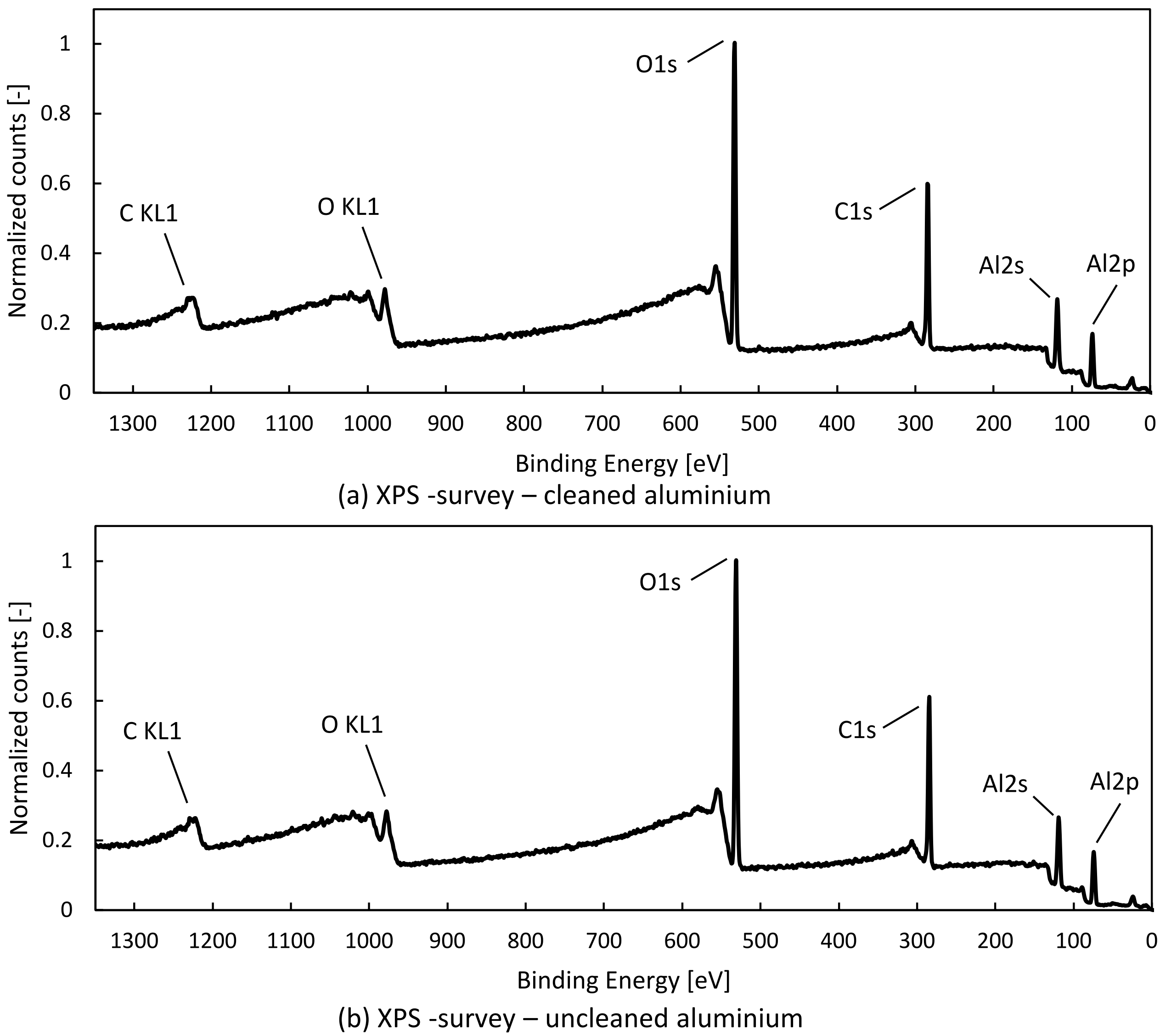

Fig. 8 XPS-survey comparison for EN AW-1050-H14 (without baseline correction)

laser-processed samples. In the cleaned and uncleaned Al samples, the O1s intensity decreases by half after about 50–70 s of sputtering time. Assuming the usual sputtering rates for aluminum oxides [62], this results in an oxide layer between ~ 2.8 and 4.5 nm. After laser processing, the O1s intensity only drops by half after around 160–200 s of sputtering time. This results in oxide layer thicknesses between ~ 9 and 12.6 nm.

3.1.2 GD-OES

If the termination criterion of 50% mass concentration compared to the plateau value proposed in accordance with the DIN ISO 14707 standard [63] is used, an oxide layer thickness of about ~ 200–250 nm is formed on the copper surface as a result of oxidation at 180 °C (Fig. 10a) for 3 days (Group T180). Since a combined layer of Cu(I) oxide (near the material) and Cu(II) oxide (near the interface) [31] is formed and the surface roughness is ~ Sa 0.3–0.35 µm, the curve of the oxygen content is not surprising. Due to the low lateral resolution (measuring point ~ 5 mm diameter) and the surface roughness of the samples, a more precise indication of the remaining oxide layer is not possible with the GD-OES under present conditions. More precise statements are only possible with highly polished and subsequently aged samples which, in turn, would only correspond to the real application to a limited extent.

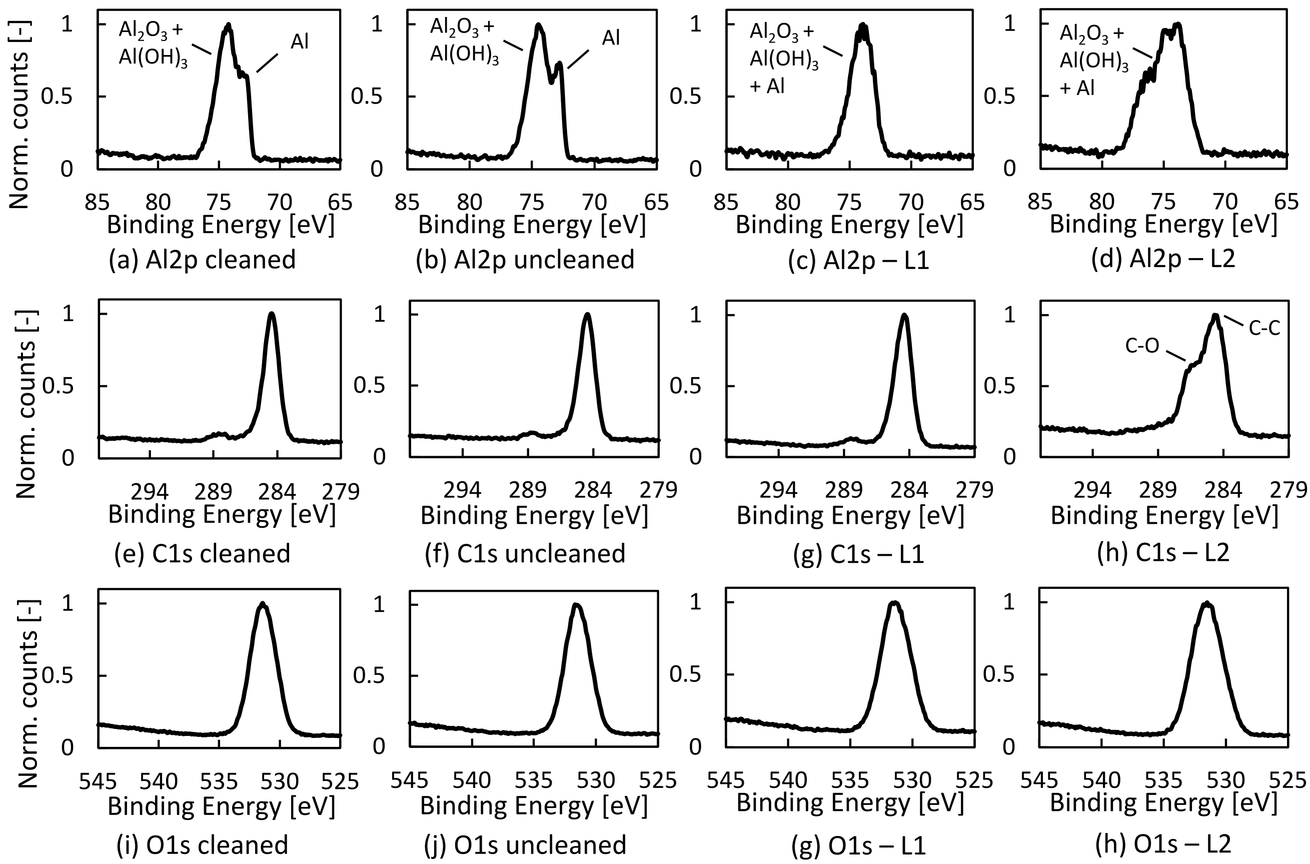

Fig. 9 Initial XPS-spectra for Al2p, C1s, and O1s of cleaned, uncleaned, and laser-processed EN AW-1050-H14 samples

In the untreated, bright state, the oxide layer thickness is < 5 nm. More precise information is also not possible here without further investigations (e.g. preparation of samples without significant technical roughness $R_a << 0.1$ µm). The GD-OES measurement of aluminium allows a similar conclusion to be drawn (Fig. 10b). The oxide layer is smaller than 5 nm.

3.1.3 Welding results (Cu-Cu)

Chemical cleaning of the sheets (*U*) leads to a reduction in the average strength compared to the uncleaned reference state (*R*), see Fig. 11a. The welding time increases (+ ~ 40 ms), the indentation depth decreases, whereby a lower scattering is achieved in each case (Fig. 11b and c).

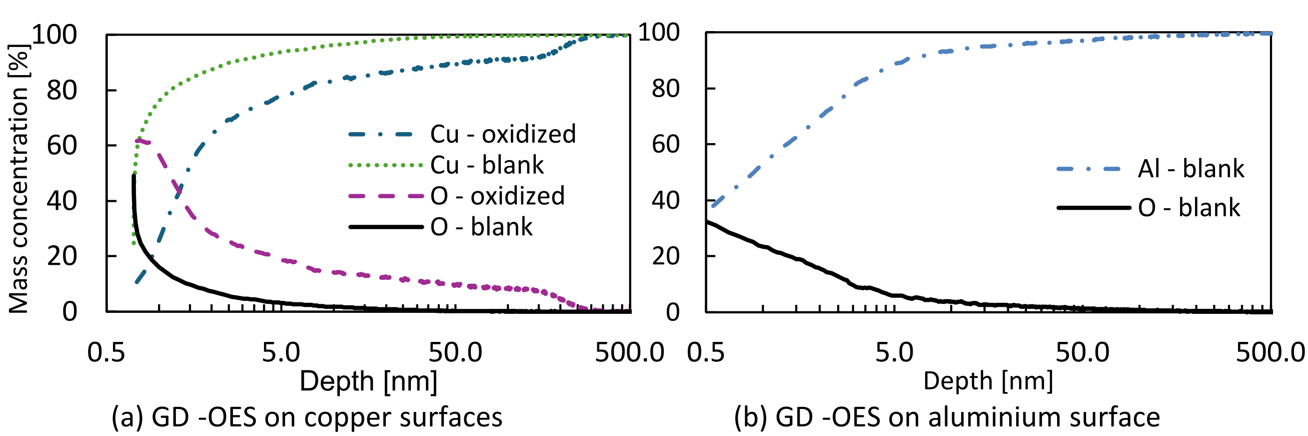

Fig. 10 GD-OES measurement results

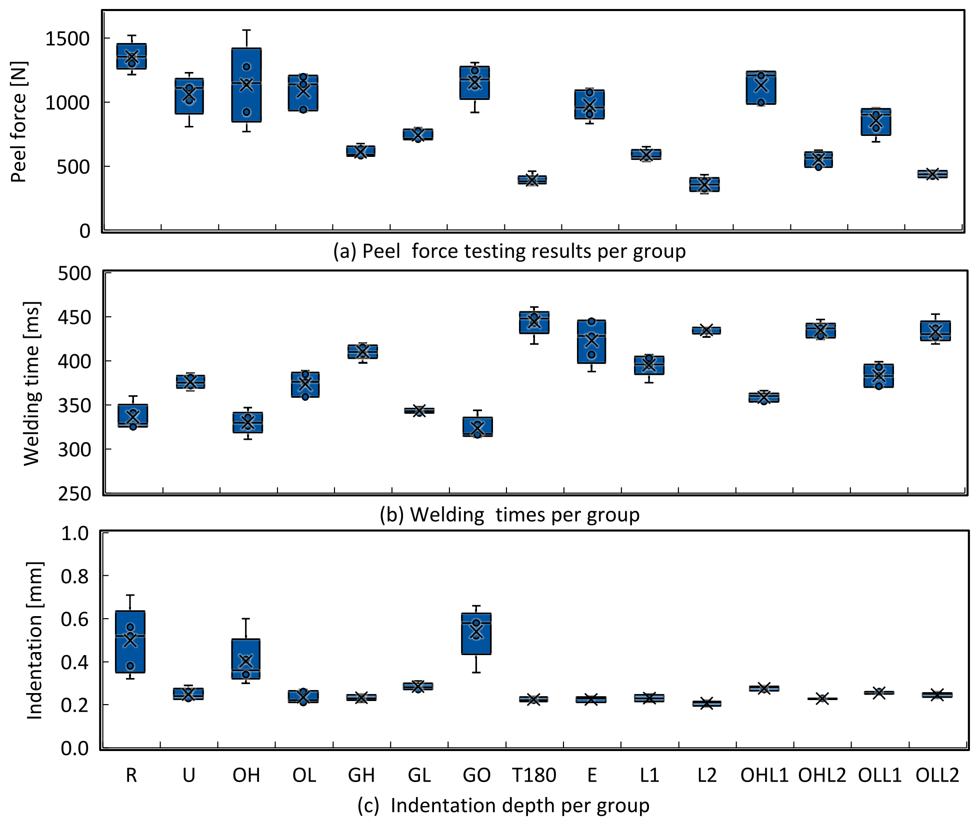

Fig. 11 Boxplots of welding results for similar copper-joints

Contamination with rolling oil (OL or OH) leads to a slightly increased joint strength compared to the cleaned state (*U*). The scattering of the peel tensile strength increases sharply with a high oil concentration (OH), while it remains almost the same with a low oil concentration (OL). The effect of the oil on the welding time is very small, but the indentation depth increases sharply. The weakest samples (T180) achieve only about 1/5 of the peel tensile force of the best samples (*R*). The reduction in strength due to increased roughness is also more than compensated for by contamination (GO) compared to a cleaned, bright condition (*U*). Nevertheless, the strength is reduced compared to the exclusively oiled or uncleaned reference condition (*R* and OH). As a result of various combinations of changes in the surface structure (roughness, waviness and oxidation) and contamination, the joint strength can therefore vary widely. All mechanical and laser-processed conditions show higher welding times than oil-contaminated conditions. The supposedly perfect, electropolished condition (*E*) does not lead to the highest joint strengths. Instead, the samples are at a comparable level to the cleaned surface condition (*U*). Laser cleaning with remelting (L2) leads to the lowest joint strengths. This condition combines the poor properties of the higher roughness (see GL and GH) and the oxidized surface (T180).

Compared to the results from Helfers et al. [12], the existing intensity of the laser system used is not sufficient to remove the contamination with diluted rolling oil without leaving any residue and at the same time not altering the surface, resulting in reduced joint strength. At the same time, it can be determined that the scatter of the joint strength can be significantly reduced by the laser treatment (L1 and L2) compared to an uncleaned state (*R*) or an oiled state (OH and OL). In order to achieve a robust overall process chain, the mentioned adaptation of laser treatment and welding process to each other is absolutely necessary. The interaction between the disturbance variables is consistent with the experience reported in the literature [7, 9, 11, 12]. Depending on the combination of boundary conditions (component geometry, welding parameters, cleanliness, tool geometry, damping, etc.), various effects can interfere with each other, leading to an increase or decrease in the joint strength that is difficult to predict based exclusively on scalar process information. In general, the number and severity of disturbances should be kept to a minimum in order to maintain a robust welding process.

The evaluation of the reference samples with regard to joint strength shows that the surface should be as oxide-free as possible for welding Cu-Cu joints, as these must be broken up and driven out at the start of the process (cleaning phase). The surface structure should also be as homogeneous as possible and have a low roughness (~ 0.2–0.4 µm). Rolling oil in the welding zone can be beneficial for joint formation. Any scattering of surface parameters (cleanliness, surface structure and oxide layer), even in combination, is characterized by a correspondingly large scattering of the joint strength.

The chemically cleaned state (*I*) exhibits similar or, in some cases, higher fluorescence than the state (OL), which indicates incomplete cleaning. A slightly higher fluorescence is identified in state (*E*) than in the thermally aged state (T180). It is not yet possible to determine whether there is really a higher residual contamination or whether the fluorescence measuring device is disturbed by the highly reflective surface (Fig. 12b).

The comparison with the calculated average resistance of the base materials (R_{base_CuCu} ~ 17 µOhm) at a 30-mm measuring length indicates that the contact resistance increases on average by ~ 1–4.5 µOhm depending on the test series. The differences in contact resistance (Fig. 13) are on average within ± 10%, which corresponds to a change of around ± 2 µOhm. There are obviously some incorrect measurements in the OL test series, as the available values are below the material resistance. A clear reduction or increase in contact resistance as a result of surface treatment, either mechanically, chemically or with the laser, cannot be determined. In fact, it can be summarized that the quasi-stationary current load test carried out and the resulting electrical resistance of ultrasonically welded Cu-Cu samples do not correlate with the joint strength.

The evaluation of the process signals (Fig. 14) once again emphasizes the importance of the combined consideration (generally sensor data fusion and intelligent evaluation) of different process information. If only individual signals are analyzed, this can lead to misinterpretation. The course of the supposedly perfect, electrochemically polished surface (*E*) is particularly interesting. The power consumption is very low, especially at the beginning. There is also very little attenuation in the horn signal. The reduction in roughness and structuring of the surface compared to the reference state (S_a ~ 0.35 µm lowered to S_a ~ 0.2 µm) as well as the complete removal of deformed layers and adsorbed substances is therefore not

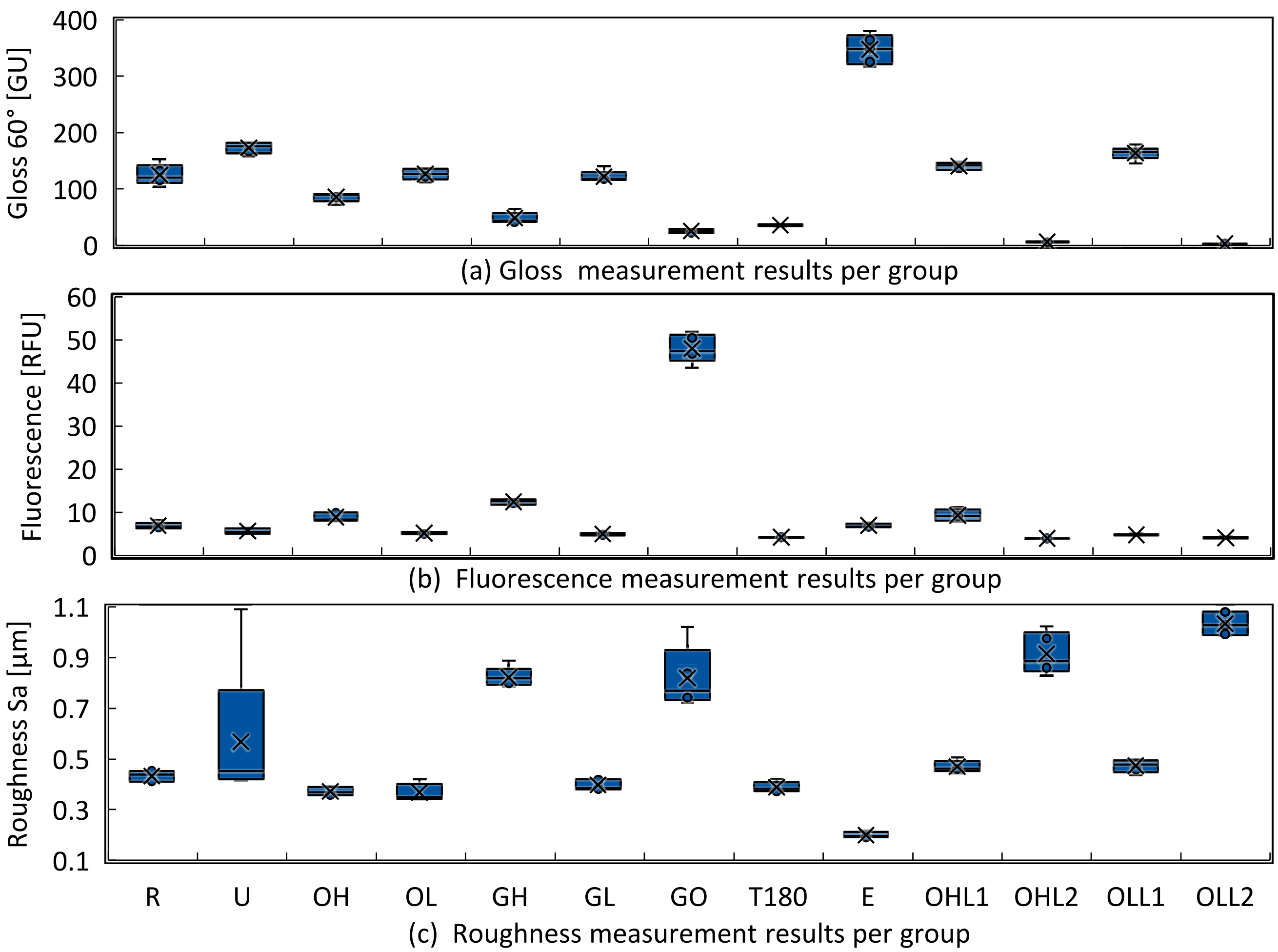

Fig. 12 Boxplots of surface data gloss, fluorescence and roughness (Cu-Cu)

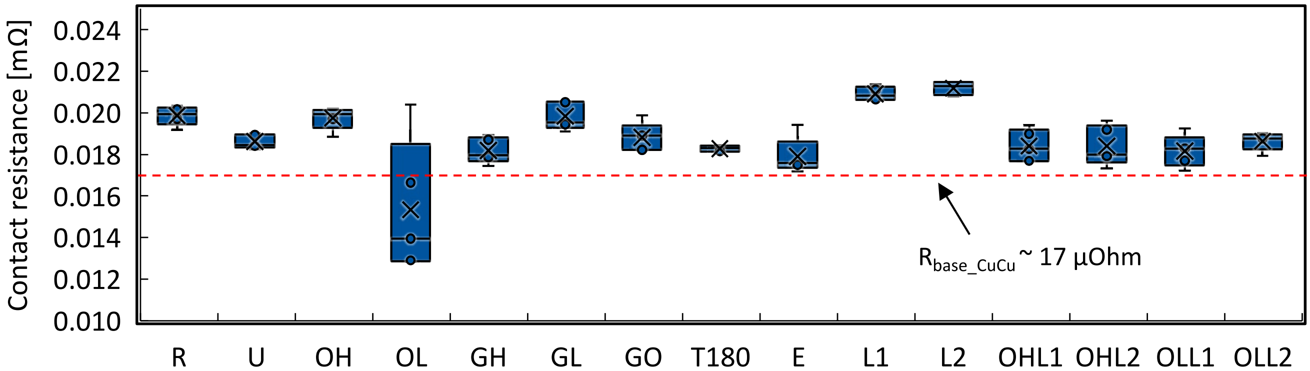

Fig. 13 Contact resistance of Cu-Cu specimen

necessary to optimise the formation of the connection. Rather, there is a state between $S_a \sim 0–0.35$ µm roughness, which is optimal for bond formation. The production of a technically almost perfectly flat surface is therefore not necessary from the USMW's process perspective. The cleaning phase already mentioned in Helfers et al. [12] is clearly evident, although in state *E* it corresponds more to a preparation phase than a cleaning phase. The mechanical processing of the roughness (GL) at the level of the reference state does not lead to a pronounced cleaning phase in the process data. At the same time, however, the mechanical processing is visible through a reduced joint strength and a lower gloss of the surface, which can be detected via the gloss measurement. This observation

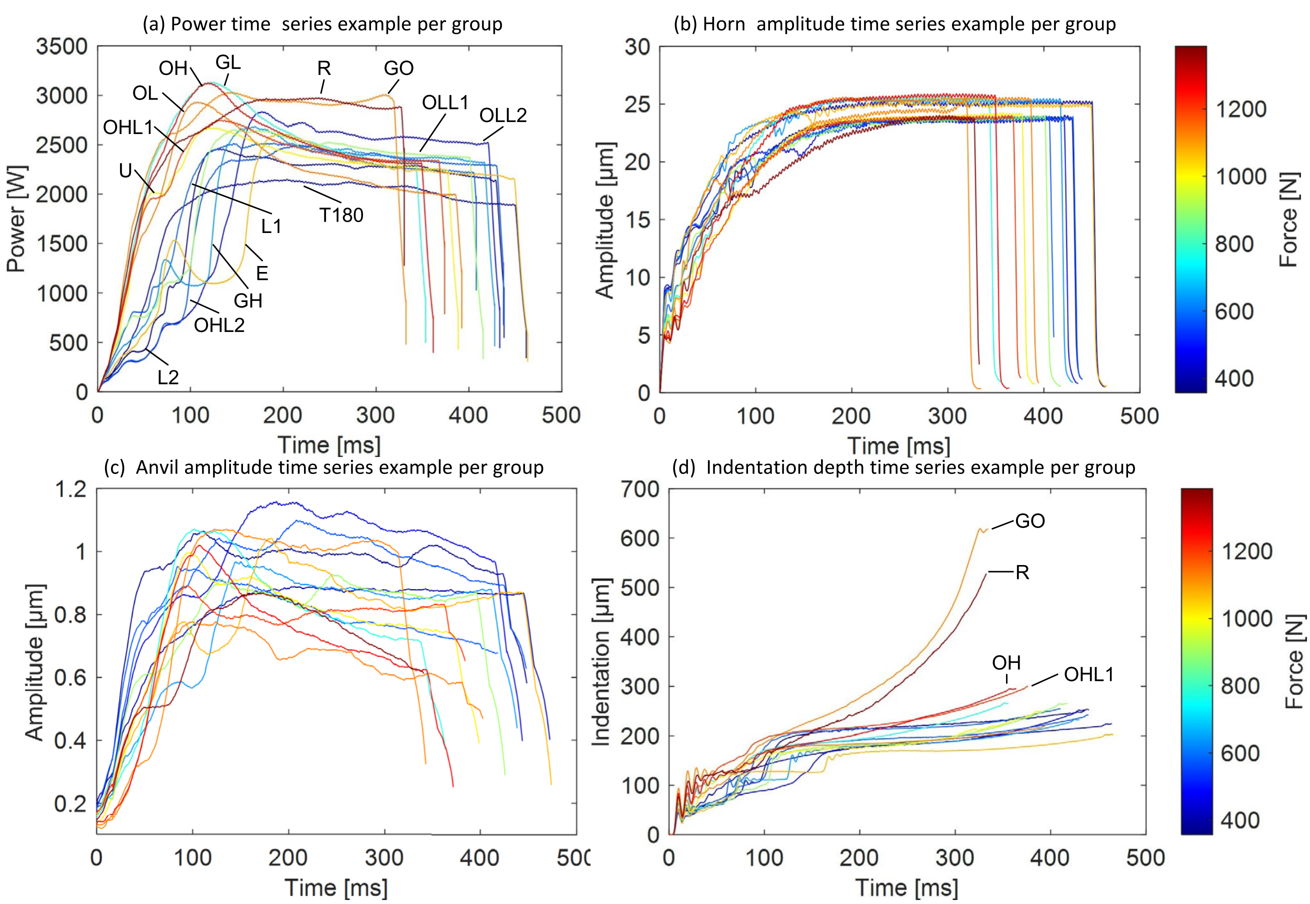

Fig. 14 Exemplary measurement curve of one sample per test series (Cu-Cu)

shows that the rolling texture from the copper manufacturing process is advantageous for joint formation. All samples with significantly higher roughness (except for GO) or thermal ageing compared to the reference condition (GH, T180, OHL2, OLL2) show a significantly lower power consumption up to around 200 ms welding time. In the GO series, the drop in power is initially compensated for by the oil. Only the correlation with the indentation depth allows a clear allocation. The amplitude curve at the horn shows that the samples with the best mechanical joint strength exhibit the greatest amplitude damping in the first 50 ms of the welding time. In addition, a final value of ~ 25 µm is reached at the horn for some of the samples and ~ 24 µm for a second group. This corresponds to an amplitude attenuation of 2 and 3 µm respectively compared to the idle oscillation. An increased amplitude attenuation in the first 50 ms welding time for the samples with the highest mechanical strength can also be seen on the anvil.

In summary, the externally recorded sensor data (power, horn, and anvil oscillation and indentation depth) is excellently suited for detecting surface-related differences during ultrasonic welding.

3.1.4 Optimization of welding parameters (Cu-Cu)

The detected loss of strength due to laser processing compared to the cleaned reference condition (*U*) can be compensated for by adjusting the welding parameters. For this purpose, a full-factorial test plan including center point (2-step, three factors) is welded with laser-processed samples (parameter L2) according to Fig. 15. The welding force is reduced by up to a third (1500 N reduced to 1000 N) in order to reduce the indentation of the horn. The amplitude is increased to the maximum possible with the existing horn in order to break up the oxide layer as quickly as possible. The welding energy is increased to double the value (800 Ws raised to 1600 Ws). As a result, the welding process becomes less economical (longer welding time 4xx to 8xx ms) and the material is subjected to greater stress (possibly higher indentation depth and annealing colors). The joint strength of the (oil-) contaminated samples remains unachievable.

The peel tensile force can be increased to > 1200 N, which is already close to the reference samples (*R*) and higher than the cleaned samples (*U*). However, this leads to a considerable formation of temper colors—the thermal-mechanical stress increases significantly. The welding time is > 800 ms and the indentation depth is at the level of the reference samples (*R*), see Fig. 16. Figure 17 shows a comparison of welded Cu-Cu joints in macrosection. In the chemically cleaned state, there is a significantly larger thinning in the area of the joint. In the detailed view, there is no longer a recognizable separation line between the sheets (Fig. 17b and c). At low laser fluence (9 J/cm^2), no separation line or accumulation of roughness peaks or oxide layers can be recognized perpendicular to the weld direction either (Fig. 17e). In a second cut along the welding direction, however, a dividing line between the sheets is already visible (Fig. 17f). The thinning, however, decreases significantly. Laser processing with high fluence (14 J/cm^2) leads to the accumulation of roughness peaks and oxide layers, which result in defects and a visible separation line between the welded sheets. The samples in Fig. 17 were welded with a

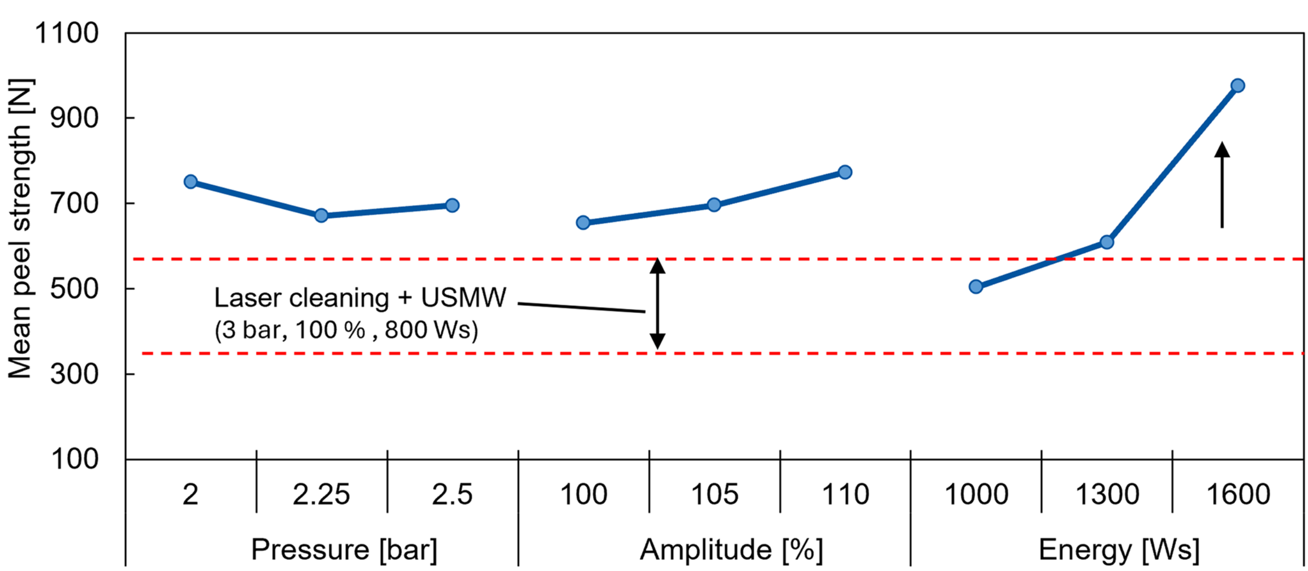

Fig. 15 Main effect diagram for welding results of full factorial experimental plan on Cu-Cu-welds

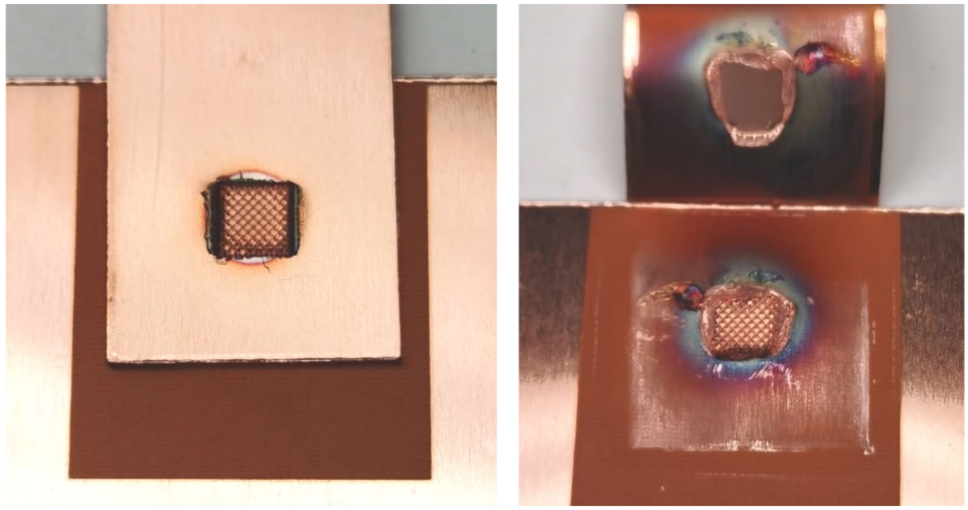

Fig. 16 Sample and fracture pattern with optimized welding parameters (2 bar, 110% amplitude and 1600 Ws)

parameter set of 2.5 bar pressure, 110% amplitude and 1300 Ws energy.

A robust overall process chain consisting of laser treatment and USMW (high joint strength and low scattering of the joint strength) is possible in principle, but has so far been accompanied by a more inefficient welding process. Laser processing of high quality, bright copper sheets as preparation for USMW is therefore only of limited use.

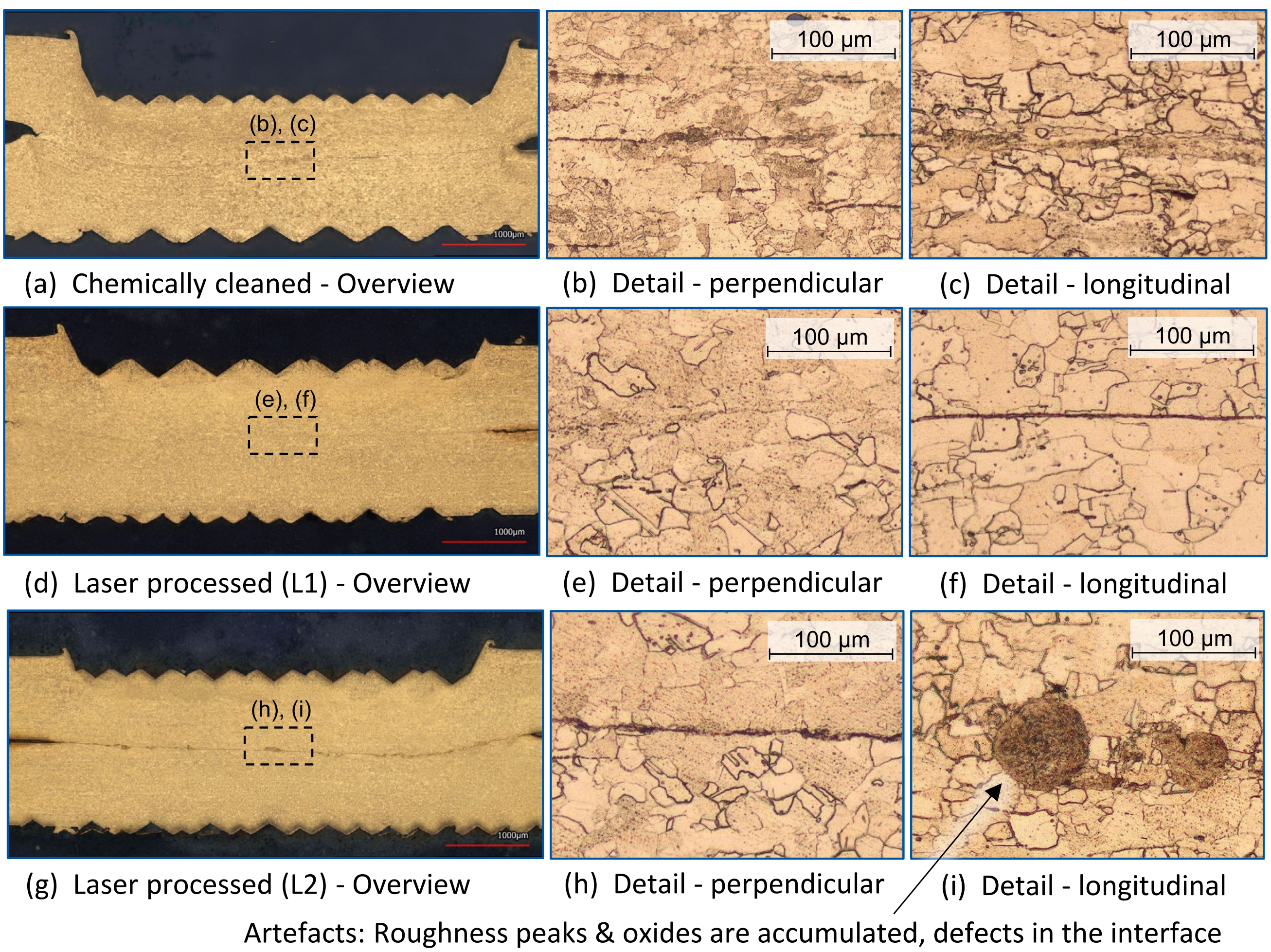

Fig. 17 Comparison of chemically cleaned and laser cleaned macrosections (Cu-Cu)

In industrial applications, however, a greater deviation in the quality of surfaces must be assumed in many cases, so that the potential of laser pre-treatment can be exploited.

3.1.5 Dissimilar Cu-Al joints

All samples are mechanically testable, as minimum strength requirements are met and there is no excessive overwelding. Chemical cleaning of the sheets (*I*) leads to a reduction in average strength compared to the uncleaned reference condition (*R*), see Fig. 18a. The welding time increases slightly (+ ~ 10 ms), the indentation depth decreases, whereby a lower scatter is achieved in each case. Contamination with rolling oil (OL or OH) leads to a significant increase in joint strength—almost doubling. The higher oil dilution (OH) already leads to a reduction in the average strength compared to OL. The scatter of the peel tensile force increases significantly. The effect of the oil on the welding time is very small, but the indentation depth increases significantly (Fig. 18b). Overall, the effect of the surface changes on the welding time (Fig. 18c) is very small (< 10% deviation compared to *R*). The mechanical processing of the surfaces leads to a reduction in the peel tensile force of ~ 20% (GL) or ~ 26% (GH) compared to the reference condition. This is only very slightly reflected in the indentation depth and in the welding time (change < 5%).

It turns out that the results of the Cu-Cu welds are largely transferable to Al-Cu dissimilar joints. The influence of oil and roughness tends to remain, even if the influence of roughness in particular is less pronounced. The previously pronounced influence of the surface condition on the welding time, on the other hand, has almost completely disappeared. The mechanical processing of the surfaces primarily leads to a greater scattering of the welding time. On the other hand, the indentation depth shows a significant influence due to the surface condition. The rolling oil (OH and OL) in particular leads to increased displacement of the aluminium in the welding area and increased notching of copper by the horn (Fig. 19a and b).

The results regarding the strength of the joint in laser-treated samples are consistent with the investigations by Watanabe et al. [5]. Laser cleaning of the aluminium with possible reoxidation has a minor influence on the joint strength. Laser treatment of copper, on the other hand, leads to a significant reduction in joint strength. As laser processing of copper simultaneously causes a change in the oxide layer and an increase in roughness, both changes result in the lowest strength of the laser-treated surfaces. In contrast, both the increase in roughness and the change in the oxide layer hardly cause any change in the joint formation in aluminium.

The correlation of the surface characteristics with the joint strength is clear, especially when viewed in combination (Fig. 20). A high gloss leads to the highest joint strengths when comparing the mechanically processed samples with the reference condition. All surface changes apart from chemical cleaning (*I*) lead to a lower gloss. The increase in roughness leads to a reduced joint strength in all cases analyzed. The correlation of XPS and fluorescence measurement of the laser-machined aluminium sample shows that there is no higher contamination with organic substances. Rather, the fluorescence measurement is disturbed by the higher roughness and altered surface structure.

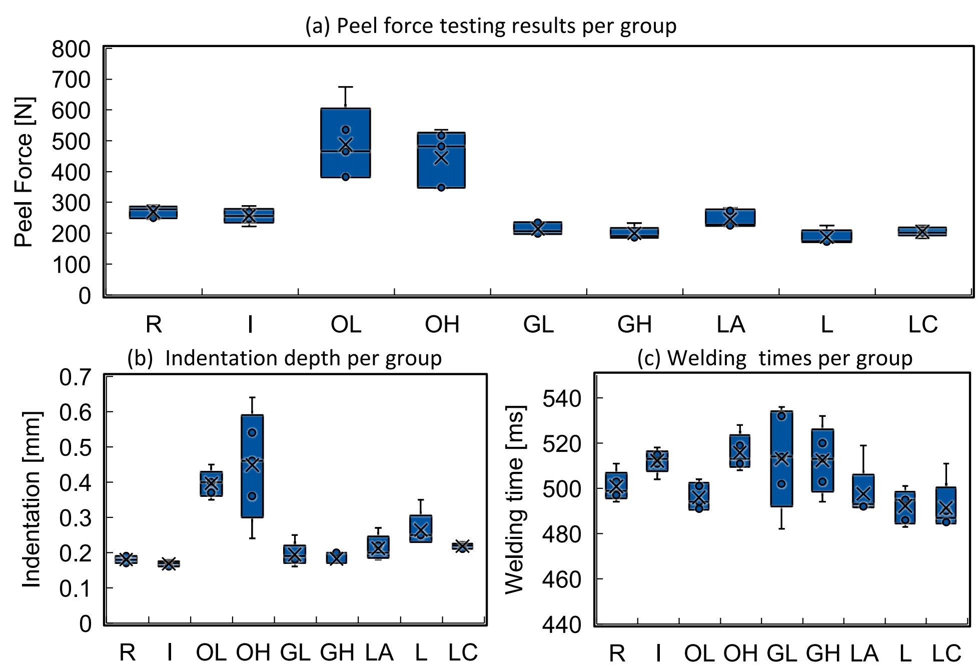

Fig. 18 Boxplots of welding results for dissimilar copper-aluminium-joints

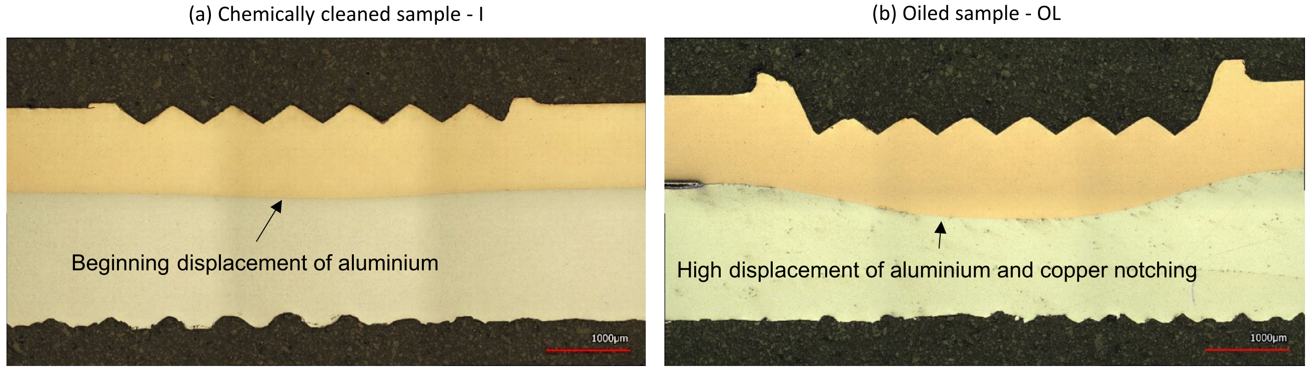

Fig. 19 Polished macrosections of Al-Cu samples

The comparison with the calculated average resistance of the base materials (R_{base_AlCu} ~ 18 µOhm) at 30 mm measuring length indicates that the contact resistance increases on average by ~ 1.5–3.5 µOhm depending on the test series. The analysis of the resistance measurement shows that laser treatment causes a higher contact resistance (up to 10%) of the welded joint compared to the reference state *R* (Fig. 21). All other surface modifications carried out also only cause slight changes in the resistance of 5–10%, which corresponds to an increase in the contact resistance of 0.5–1 µOhm. As

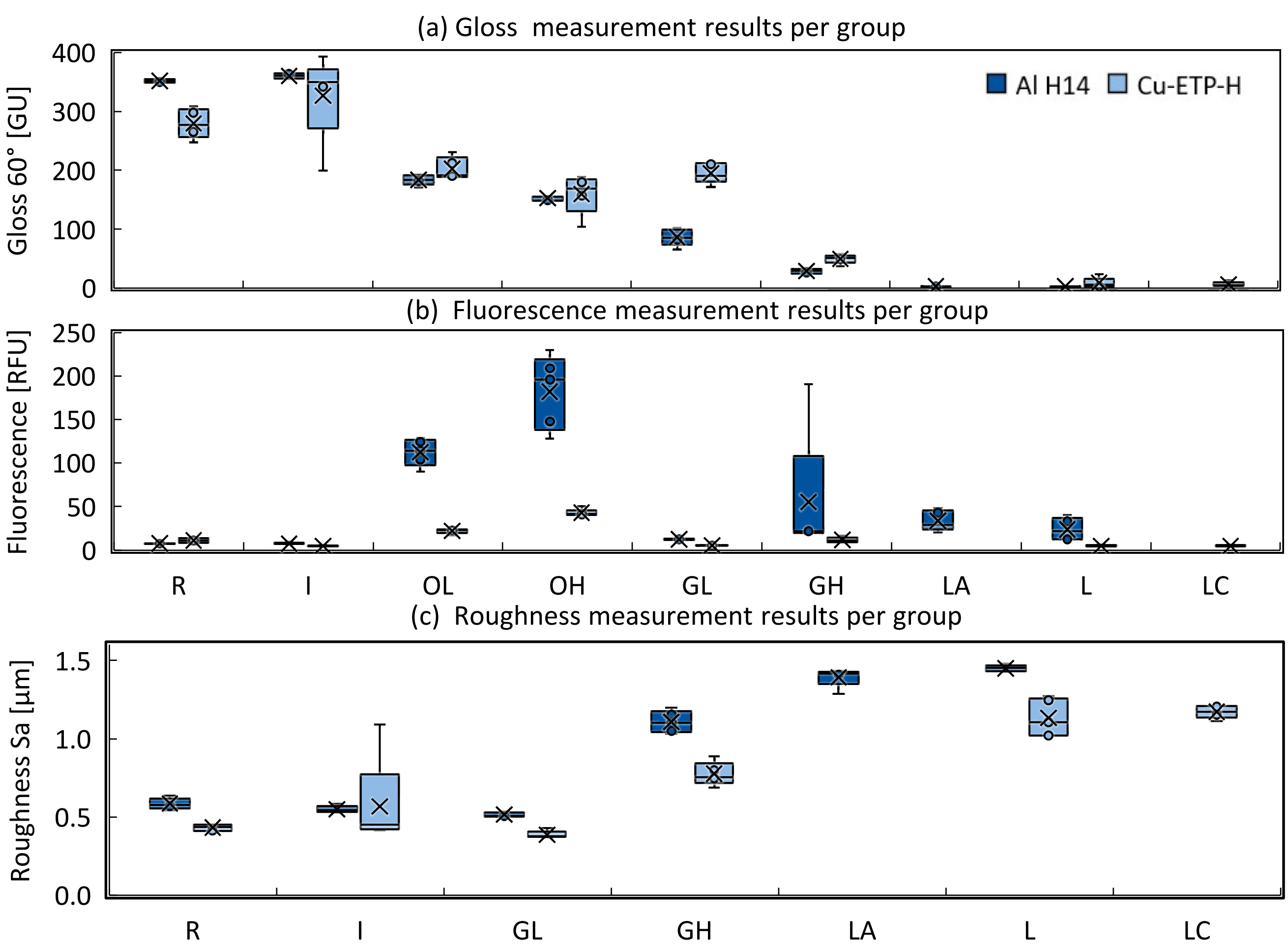

Fig. 20 Boxplots of surface data gloss, fluorescence, and roughness

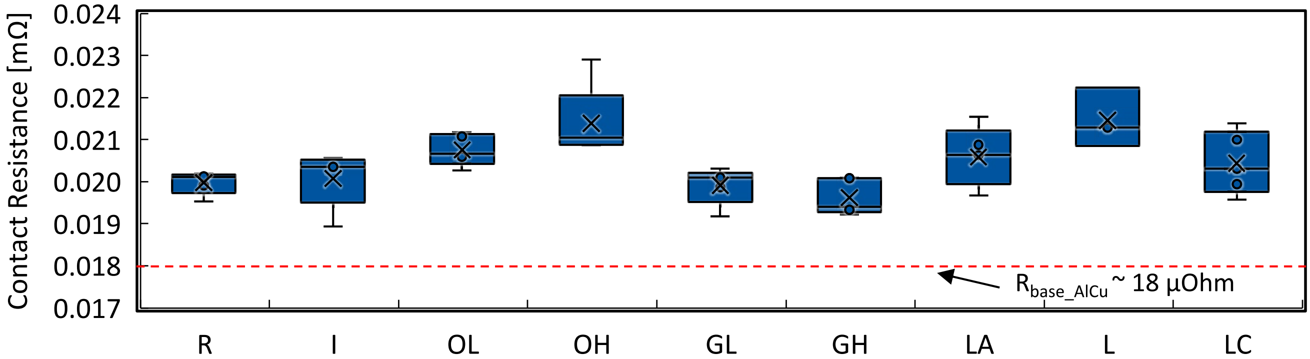

Fig. 21 Contact resistance of Al-Cu specimen

even a shift in the measuring distance of around 1–2 mm can cause a similar change, the significance of these results still needs to be verified. As with the Cu-Cu specimens, no direct correlation between quasi-stationary current load test and mechanical joint strength can be determined here either.

The mechanical joint strength can be correlated with the process behavior (Fig. 22). The oil-contaminated specimen (OL and OH) shows the greatest change compared to the reference state. At the beginning of the process (cleaning phase), there is a clearly flattened power curve. The effect of the oil on the electrical power used in the process therefore differs significantly from the Cu-Cu application. However, the effect on joint strength and indentation depth is comparable, with both increasing. The oiled samples also show the

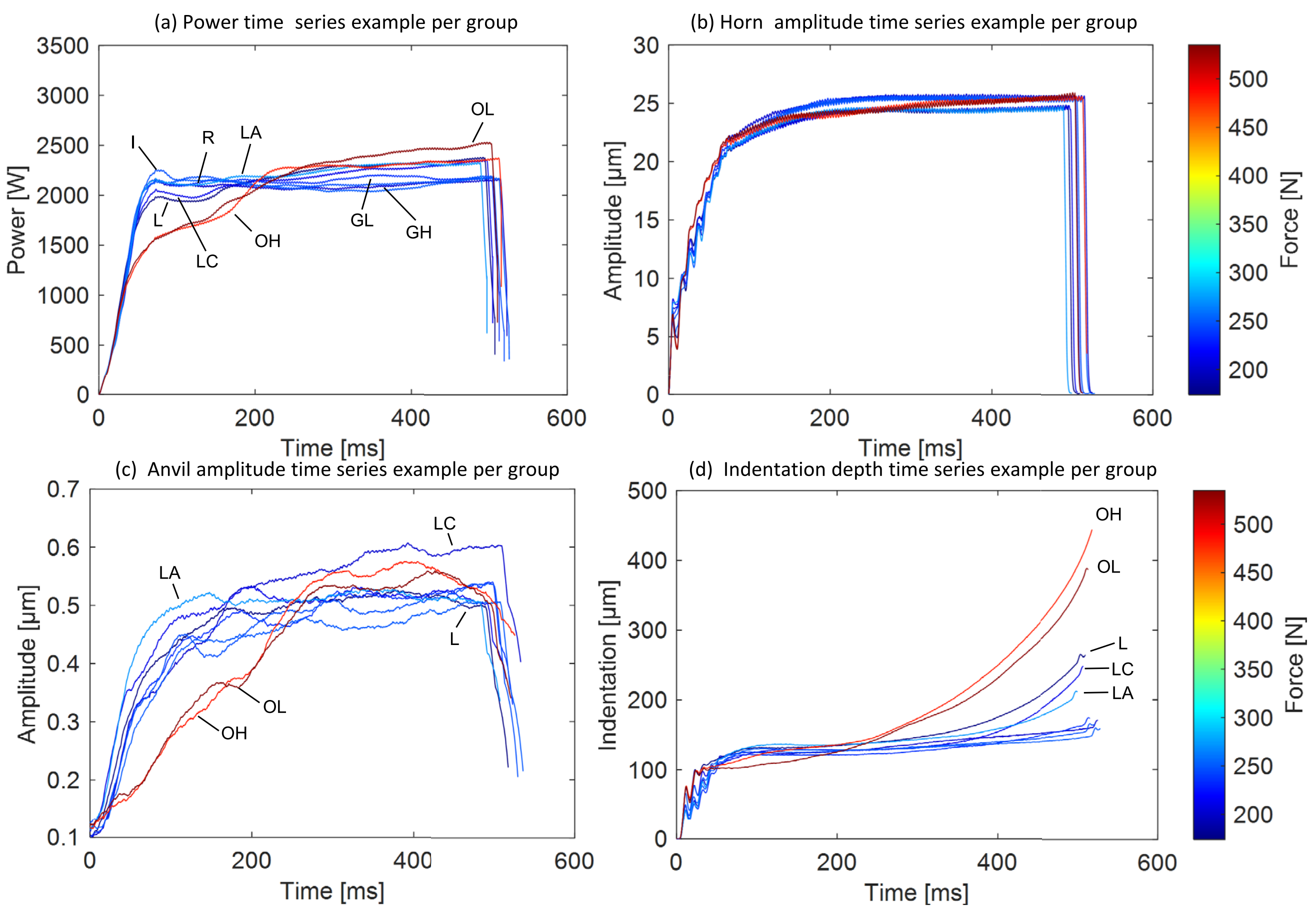

Fig. 22 Exemplary measurement curve of one sample per test series (Al-Cu)

greatest indentation depth. Up to around 200 ms welding time, the amplitude at the anvil is dampened in the oiled samples compared to the other samples. The cleaning phase, due to an increase in roughness (GL and GH), which is also clearly present in the Cu-Cu samples, is not present in the Al-Cu samples. The minor influence of the altered surface structure on the joint strength is also reflected in the process data. All laser-processed samples (L, LA and LC) are characterized by a greater indentation depth than the reference samples without oil contamination (R, *I*, GL, GH). However, this is not reflected in a higher bond strength. The oscillation amplitude at the horn is damped by about 1–1.5 µm for the laser-processed samples compared to all other samples starting from about 300 ms welding time.

To summarize, it can be said that the

Al-Cu welding process reacts similarly sensitively to contamination with oil as the Cu-Cu welding process, both in terms of the mechanical-technological properties and the process data. The influence of roughness and oxide layer tends to remain, but is significantly less pronounced. The Al-Cu welding process thus reacts relatively robustly to a change in roughness in the considered area.

4 Summary

In this study, the influences of industry-relevant surface-related influences on the USMW were examined for the first time in combination, and an optimal condition was defined based on this. In addition, it was shown in detail why laser processing of the copper surfaces leads to reduced joint strength. The development of a robust overall process chain consisting of laser cleaning and ultrasonic welding (Cu-Cu) is possible but has so far been associated with lower efficiency. The ultrasonic welding process takes significantly longer (up to 100% longer), and more energy has to be invested in the process (up to 100% more) to compensate for the loss of strength. The reasons for the reduced joint strength as a result of laser treatment (growing oxide layer and increased roughness) were also described in detail. The investigations show that the delivery condition of the copper sheet is already excellent. However, this cannot always be assumed in the application, as shown by the respective disturbance variables. The development of a robust overall process chain consisting of environmentally friendly laser cleaning and ultrasonic welding is therefore still desirable, particularly with regard to the reconditioning of components for USMW.

The transferability of basic phenomena from ultrasonically welded Cu-Cu joints to Al-Cu joints is largely given. The comparison of the process data also emphasizes that the respective relevance and significance of the parameters (e.g., indentation or power) depend on the material combination. This should be taken into account in particular when developing process monitoring strategies. The gloss and fluorescence measurements are valuable methods for characterizing the weldability of specimen surfaces with USMW and for supporting quality prediction.

In addition, the following statements can be made about a reference condition and laser pre-treatment for USMW:

- Any increase in the (micro) roughness or (micro) structuring of the surface compared to the reference state shown has a negative effect on the joint formation. Roughness peaks must be removed before the weld is formed. Rolling grooves can improve joint formation.
- Oil contamination between the parts to be welded can improve joint formation but also lead to over-welding and component pinching. A thicker oxide layer reduces the joint strength.
- Laser structuring with a short-pulsed laser (Gaussian profile) and existing reoxidation is not helpful for joint formation during ultrasonic metal welding. However, precise adjustment of the laser processing parameters to the existing contamination without melting can increase the robustness of the USMW process.
- During single-stage laser processing of aluminium and copper in nanosecond range with remelting (intensity above the coupling limit) and under atmospheric conditions, reoxidation cannot be avoided so far.

4.1 Outlook

The basic knowledge presented on the influence of various surface properties and conditions provides users with an understanding of the expected process changes during USMW of copper and Al-Cu dissimilar joints. In addition, the following further investigations can be carried out on this basis:

- Detailed analysis and correlation of XPS measurements carried out on processed copper and aluminum sheets with the influence on the suitability for USMW.
- Optimization of the surface treatment of copper and aluminium surfaces with short-pulsed and ultra-short-pulsed laser irradiation (roughness, oxidation and efficiency) with regard to the requirements of USMW—e.g. with a top-hat beam profile or different laser wavelength.
- Detailed investigation of (two-step) laser processing in terms of oxide formation and micro-roughness in combination with resulting effects on USMW.
- Integration of combined surface (e.g. fluorescence and gloss) and USMW condition monitoring, which enables robust welding process prediction.

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Data availability Data will be made available on reasonable request by the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

References

1. Wodara J, Adam T (2004) Ultraschallfügen und -trennen: Band 1 der Grundlagen der Fügetechnik. Düsseldorf: DVS Verlag. ISBN: 3-87155-212-7
2. Balz I, Raad EA, Rosenthal E et al (2020) Process monitoring of ultrasonic metal welding of battery tabs using external sensor data. J Adv Join Process 1:100005. https://doi.org/10.1016/j.jajp.2020.100005
3. Shawn Lee S, Hyung Kim T, Jack Hu S, Cai WW, Abell JA, Li J (2013) Characterization of joint quality in ultrasonic welding of battery tabs. J Manuf Sci Eng. https://doi.org/10.1115/1.4023364
4. Balz I. Prozessanalyse der thermo-mechanischen Vorgänge während der Verbindungsbildung beim Metall-Ultraschallschweißen // Analysis of thermo-mechanical processes during the bond formation in ultrasonic metal welding. 1. Auflage. Düren: Shaker 2020. https://doi.org/10.18154/RWTH-2020-08744
5. Watanabe T, Yanagisawa A, Konuma S, Yoneda A, Ohashi O. Effect of oxide film on bond strength of ultrasonically welded joints, and welding process. Study of ultrasonic welding of dissimilar metals. Welding International 1999; 13(12): 936–44 https://doi.org/10.1080/09507119909452077
6. Lindamood LR, Matheny MP, Graff KF. Chapter 10 - Ultrasonic welding of metals. In: Gallego-Juárez JA, Graff KF, Lucas M, editors. Power ultrasonics: applications of high-intensity ultrasound. Second edition. Oxford: Woodhead Publishing 2022; 163–204 https://doi.org/10.1016/B978-0-12-820254-8.00012-9
7. Müller FW, Mirz C, Weil S, Schiebahn A, Corves B, Reisgen U (2023) Weld quality characterization by vibration analysis for ultrasonic metal welding processes. Journal of Advanced Joining Processes 8:100149. https://doi.org/10.1016/j.jajp.2023.100149
8. Müller FW, Liu J, Schiebahn A, Reisgen U (2024) Influence of workpiece geometry and natural frequencies on ultrasonic metal welding. Proc Inst Mech Eng Part L J Mater Des Appl. https://doi.org/10.1177/14644207241245431
9. Müller FW, Schiebahn A (2019) Reisgen u untersuchungen zum störeinfluss von werkstoff- und oberflächeneigenschaften auf cu-cu metall-Ultraschallschweißverbindungen. METALL, 73. Jahrgang: Seite 463–467
10. Nunes R, Faes K, de Meester S, de Waele W, Kubit A (2022) Influence of welding parameters and surface preparation on thin copper–copper sheets welded by ultrasonic welding process. Int J Adv Manuf Technol 123(1–2):373–388. https://doi.org/10.1007/s00170-022-10164-9
11. Harthoorn JL. Ultrasonic metal welding. Dissertation.Technische Hogeschool Eindhoven 1978. https://doi.org/10.6100/IR161561
12. Helfers E, Müller FW, Schiebahn A, Reisgen U (2024) Influence of surface condition of copper sheets on ultrasonic metal welding. J Adv Join Process 9:100204. https://doi.org/10.1016/j.jajp.2024.100204
13. Müller FW, Chen C-Y, Schiebahn A, Reisgen U (2023) Application of electrical power measurements for process monitoring in ultrasonic metal welding. Weld World 67(2):395–415. https://doi.org/10.1007/s40194-022-01428-9
14. Adam T (1999) Ultraschallschweißen ausgewählter Aluminiumlegierungen mit erhöhter Festigkeit. Dissertation, Institut für Füge- und Strahltechnik, Otto-von-Guericke-Universität
15. Mostafavi S. Multiscale modeling and experimental study of ultrasonic metal welding. RWTH Aachen University 2022. https://doi.org/10.18154/RWTH-2023-01629
16. Lee SS, Shao C, Kim TH*, et al.* Characterization of ultrasonic metal welding by correlating online sensor signals with weld attributes. Transactions of the ASME, Journal of Manufacturing Science and Engineering 2014; 136(5): 051019/1–10 https://doi.org/10.1115/MSEC2014-4139
17. Daniels H (1965) Ultrasonic welding. Ultrasonics 3(4):190–6. https://doi.org/10.1016/0041-624X(65)90169-1
18. Becker M. Energieeffizientes Punktschweißen von Aluminium/Stahl-Strukturen durch Leistungsultraschall 2022. https://doi.org/10.6094/UNIFR/231474
19. Fachverband Industrielle Teilereinigung e.V., editor. FiT-Richtlinie: Filmische Verunreinigungen beherrschen, 2021.
20. Schneider-Ramelow M, Rudolf F (2010) Lieferempfehlung für Leiterplatten PLUS 12:556–559
21. Eacock F, Unger A, Eichwald P*, et al.* Effect of different oxide layers on the ultrasonic copper wire bond process: 2111–8. https://doi.org/10.1109/ECTC.2016.91
22. Ostermann F. Anwendungstechnologie Aluminium. 3., neu bearbeitete Auflage. Berlin, Heidelberg: Springer Vieweg 2014. https://doi.org/10.1007/978-3-662-43807-7
23. Kästle C. Qualifizierung der Kupfer-Drahtbondtechnologie für integrierte Leistungsmodule in harschen Umgebungsbedingungen. FAU University Press 2019. https://doi.org/10.25593/978-3-96147-146-1
24. DIN EN ISO 2813:2015–02: Paints and varnishes - determination of gloss value at 20°, 60° and 85° (ISO 2813:2014); German version EN ISO 2813:2014. Berlin: Beuth Verlag GmbH. https://doi.org/10.31030/2146718
25. Lohse A, Schümann L (2013) Sauberkeitskontrolle direkt am Prozess. J Oberfl Tech 53(16):56–7. https://doi.org/10.1365/s35144-013-0846-6

26. Stiedl J, Green S, Chassé T, Rebner K (2019) Characterization of oxide layers on technical copper material using ultraviolet visible (UV-Vis) spectroscopy as a rapid on-line analysis tool. Appl Spectrosc 73(1):59–66. https://doi.org/10.1177/0003702818797959
27. Al Ktash M, Stefanakis M, Englert T et al (2021) UV hyperspectral imaging as process analytical tool for the characterization of oxide layers and copper states on direct bonded copper. Sensors. https://doi.org/10.3390/s21217332
28. Godard HP (1967) Oxide film growth over five years on some aluminum sheet alloys in air of varying humidity at room temperature. J Electrochem Soc 114(4):354. https://doi.org/10.1149/1.2426593
29. Zähr J, Oswald S, Türpe M, Ullrich HJ, Füssel U (2012) Characterisation of oxide and hydroxide layers on technical aluminum materials using XPS. Vacuum 86(9):1216–9. https://doi.org/10.1016/j.vacuum.2011.04.004
30. Hunter MS, Fowle P (1956) Natural and thermally formed oxide films on aluminum. J Electrochem Soc 103(9):482. https://doi.org/10.1149/1.2430389
31. Leygraf C, Odnevall Wallinder I, Tidblad J, Graedel T. Atmospheric corrosion. Second edition. Hoboken, New Jersey: Wiley 2016. https://doi.org/10.1002/9781118762134
32. Berg J. Kupfer als Katalysator für Partialoxidationen. Darmstadt, Techn. Univ., Diss., 2009. https://doi.org/20104.
33. Rhodin TN (1950) Low temperature oxidation of copper. I. Physical mechanism1a. J Am Chem Soc. https://doi.org/10.1021/ja01167a079
34. Cabrera N, Mott NF (1949) Theory of the oxidation of metals. Rep Prog Phys 12(1):163–84. https://doi.org/10.1088/0034-4885/12/1/308
35. Pinnel MR, Tompkins HG, Heath DE (1979) Oxidation of copper in controlled clean air and standard laboratory air at 50°C to 150°C. Appl Surf Sci 2(4):558–77. https://doi.org/10.1016/0378-5963(79)90047-3
36. Copper Development Association. Electronic connector design guide, 2023 [cited 2023 October 25] Available from: URL: https://copper.org/applications/industrial/DesignGuide/.
37. Wiederholt W (1964) Die atmosphärische Korrosion von Kupfer und Kupferlegierungen. Mater Corrosion 15(8):633–44. https://doi.org/10.1002/maco.19640150804
38. DIN 8592:2003–09, Manufacturing processes cleaning - classification, subdivision, terms and definitions. Berlin: Beuth Verlag GmbH .https://doi.org/10.31030/9500682
39. Reinigungstechnik F-A (2013) Markt- und Trendanalyse in der Industriellen Teilereinigung 2012. Frauenhofer-Allianz Reinigungstechnik, Berlin
40. Satta A, Shamiryan D, Baklanov MR et al (2003) The removal of copper oxides by ethyl alcohol monitored *in situ* by spectroscopic ellipsometry. J Electrochem Soc 150(5):G300. https://doi.org/10.1149/1.1564108
41. Lee JM, Watkins KG (2000) In-process monitoring techniques for laser cleaning. Opt Lasers Eng 34(4):429–42. https://doi.org/10.1016/S0143-8166(00)00073-7
42. Mandolfino C, Lertora E, Genna S, Leone C, Gambaro C (2015) Effect of laser and plasma surface cleaning on mechanical properties of adhesive bonded joints. Procedia CIRP 33:458–63. https://doi.org/10.1016/j.procir.2015.06.054
43. Büchter E. A Green Way to Clean - Laser Cleaning. LTJ 2012; 9(5): 36–8. https://doi.org/10.1002/latj.201290068
44. Lee JM, Watkins KG, Steen WM (2001) Characterization of laser cleaning of copper for soldering processes. J Manuf Sci Eng 123(3):521–7. https://doi.org/10.1115/1.1344897
45. Lee JM, Watkins K (2000) Laser removal of oxides and particles from copper surfaces for microelectronic fabrication. Opt Express 7(2):68–76. https://doi.org/10.1364/oe.7.000068
46. Seo C, Ahn D, Kim D (2015) Removal of oxides from copper surface using femtosecond and nanosecond pulsed lasers. Appl Surf Sci 349:361–7. https://doi.org/10.1016/j.apsusc.2015.05.011
47. Hügel H, Dausinger F. 1.1 Fundamentals of laser-induced processes. In: Poprawe R, Weber H, Herziger G, editors. Laser Applications. Berlin/Heidelberg: Springer-Verlag 2004; 25–68 https://doi.org/10.1007/10877768_2
48. Hummel M, Schöler C, Häusler A, Gillner A, Poprawe R (2020) New approaches on laser micro welding of copper by using a laser beam source with a wavelength of 450 nm. J Adv Join Process 1:100012. https://doi.org/10.1016/j.jajp.2020.100012
49. Helm J. Prozessstabilität und Prozesseffizienz beim Laserstrahlfügen von hoch reflektiven Kupferwerkstoffen. Dissertation 2022. https://doi.org/10.18154/RWTH-2022-09483
50. Pöthig P, Grätzel M, Bergmann JP (2023) Influence of different surface conditions on mechanical properties during ultrasonic welding of aluminum wire strands and copper terminals. Weld World 67(6):1427–36. https://doi.org/10.1007/s40194-023-01490-x
51. Ao SS, Li CJ, Zhang W, Oliveira JP, Zeng Z, Luo Z (2022) Effect of laser surfacing on the microstructure and mechanical properties of ultrasonic welded NiTi joints. J Manuf Sci Eng. https://doi.org/10.1115/1.4051330
52. Bubert H, Grallath E, Quentmeier A, Wielunski M, Borucki L (1995) Comparative investigation on copper oxides by depth profiling using XPS, RBS and GDOES. Anal Bioanal Chem 353(3–4):456–63. https://doi.org/10.1007/s0021653530456
53. Greczynski G, Hultman L (2022) A step-by-step guide to perform x-ray photoelectron spectroscopy. J Appl Phys. https://doi.org/10.1063/5.0086359
54. DIN EN 13599:2014–12, Copper and copper alloys - copper plate, sheet and strip for electrical purposes; German version EN 13599:2014. Berlin: Beuth Verlag GmbH. https://doi.org/10.31030/2251682
55. DIN EN ISO 6892–1:2020–06, Metallic materials - tensile testing - Part 1: Method of test at room temperature (ISO 6892–1:2019); German version EN ISO 6892–1:2019. Berlin: DIN Media GmbH. https://doi.org/10.31030/3132591
56. DIN EN ISO 25178–2:2023–09, Geometrical product specifications (GPS) - surface texture: Areal - Part 2: Terms, definitions and surface texture parameters (ISO 25178–2:2021); German version EN ISO 25178–2:2022. Berlin: DIN Media GmbH. https://doi.org/10.31030/3294021
57. Simpson R. The depth profiling of organic and inorganic materials using Ar+ and Ar+n ion beams. Dissertation,University of Surrey 2018.
58. Spittel M, Spittel T. Mechanical and physical properties of alloys and metals. In: Spittel M, Spittel T, Warlimont H, editors. Part 3: Non-ferrous alloys - heavy metals. Berlin, Heidelberg: Springer Berlin Heidelberg 2016; 87–113. https://doi.org/10.1007/978-3-642-14174-4_4
59. Spittel M, Spittel T, Warlimont H, editors. Part 3: Non-ferrous alloys - heavy metals. Berlin, Heidelberg: Springer Berlin Heidelberg, 2016. https://doi.org/10.1007/978-3-642-14174-4
60. Biesinger MC. Advanced analysis of copper X-ray photoelectron spectra. Surface & Interface Analysis 2017; 49(13): 1325–34. https://doi.org/10.1002/sia.6239
61. Justin Gorham. NIST X-ray photoelectron spectroscopy database - SRD 20. National Institute of Standards and Technology, 2012: 2012. https://doi.org/10.18434/T4T88K
62. Araki S, Tohma H, Japan SPGoSASo. Effects of charging and crystallinity on sputter-etching rate of Al2O3 single crystal. JSA 2002; 9(3): 451–4. https://doi.org/10.1384/jsa.9.451
63. DIN ISO 14707:2023–05, Surface chemical analysis - glow discharge optical emission spectrometry (GD-OES) - introduction to use (ISO 14707:2021). Berlin: Beuth Verlag GmbH https://doi.org/10.31030/3401905